

PRECOMPOSITION ON CONTINUOUS ALTERNATING MAPS IS NOT ANALYTIC IN POSITIVE CHARACTERISTIC

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ABSTRACT. For normed spaces E, E', F over a nontrivially normed field K and a finite set ι , precomposition $f \mapsto (m \mapsto m \circ (f, \dots, f))$ sends continuous linear maps $E \rightarrow E'$ to continuous linear maps between spaces of continuous alternating ι -linear maps. This map is C^n for every $n \in \mathbb{N} \cup \{\infty\}$ in every characteristic [7], and it is analytic when $|\iota|!$ is invertible in K . We show that this condition is also necessary: for every nontrivially normed field K of characteristic $p > 0$ and every $k \geq p$ there are K -Banach spaces E and F such that, with $E' = E$ and $|\iota| = k$, precomposition is analytic at no point. So precomposition is analytic for all Banach spaces over K if and only if $|\iota|! \neq 0$ in K . The target F admits no equivalent ultrametric norm. The heart of the proof is an algebraic theorem over finite fields: when $k! = 0$, the family of coordinatewise multiplication operators on sequences has no lift in degree k that satisfies the diagonal identity and has uniformly bounded support. The algebraic theorem is proved with Ramsey's theorem and an exact cancellation of sign-weighted clusters of indices, and is combined with a Banach completion of the exterior power over a Laurent series field, whose constant-coefficient map has bounded support on bounded sets, and with an ascent to an arbitrary base field through Ingleton's Hahn–Banach theorem.

1. INTRODUCTION

Let K be a nontrivially normed field. For normed spaces E, E', F over K and a finite set ι we write $\mathcal{L}(E, E')$ for the space of continuous linear maps with the operator norm, and $\text{Alt}^\iota(E; F)$ for the normed space of continuous alternating ι -linear maps $E^\iota \rightarrow F$. *Alternating* is meant in the strong sense: the map vanishes whenever two of its arguments coincide. We study the precomposition map

$$(1.1) \quad Q = Q_{E, E', F}^K: \mathcal{L}(E, E') \rightarrow \mathcal{L}(\text{Alt}^\iota(E'; F), \text{Alt}^\iota(E; F)), \quad Q(f)(m) = m \circ (f, \dots, f).$$

In Mathlib it is `ContinuousAlternatingMap.compContinuousLinearMapCLM`.

Motivation. In Mathlib, manifolds and vector bundles are modeled on normed spaces (of possibly infinite dimension) over an arbitrary nontrivially normed field, and their regularity is an element $n \in \mathbb{N} \cup \{\infty, \omega\}$, where C^ω means analytic. Differential forms need the bundle of continuous alternating maps between two C^n vector bundles to be a C^n vector bundle again. Its transition functions are those of the given bundles with Alt^ι applied, so this comes down to the regularity of Q ; compare Lang's notion of a functor of class C^p [4, Ch. III, §4]. The following is known.

The result, its proof and this write-up were produced with Anthropic's Claude (in Claude Code). The proofs were adversarially audited by independent AI agents (separate Claude instances), but neither the proofs nor this text has yet been reviewed by a human expert. Unlike the characteristic-2 result of [6], Theorem 1 is *not* machine-checked in Lean. Of the results of this paper, only the lift criterion (Proposition 6) and the positive direction of Corollary 2 are machine-checked, together with a few auxiliary lemmas and the quoted results of [7] and Theorem A (§7); the characteristic-2 counterexample of [6] (a different construction, over complete fields that are not spherically complete) is machine-checked as well. See §8.

- Over every K , with no hypothesis on the characteristic or the dimension, Q is of class C^n for every $n \in \mathbb{N} \cup \{\infty\}$, in a draft pull request to Mathlib [7].
- If $|\iota|!$ is invertible in K then Q is analytic, by polarization. In pull requests to Mathlib, this is done in characteristic 0 by Gouëzel [1], who uses it to construct the bundle for every n in characteristic 0 [2]. A proof in every characteristic is recalled in §6.
- If K is ultrametric and F is ultrametric and spherically complete, then Q is analytic for all normed E, E' . A special case, in the setting of [6], is stated without proof in [6, Remark 16]; a proof of the general statement has been formalized in Lean, in an unpublished development (§8); we quote it and do not prove it. We call it Theorem A (§7). It is not used in the proofs of Theorem 1 and Corollary 2, only in §7.
- Over every complete ultrametric field of characteristic 2 that is not spherically complete, Q is *not* analytic at 0 for $E = E' = \ell^\infty$, $|\iota| = 2$ and a suitable ultrametric Banach space F [6]. This is machine-checked in Lean under the hypothesis (K2) of [6], a nested sequence of closed balls with empty intersection, which is equivalent to non-spherical completeness by an elementary argument (Remark 29). In characteristic $p \geq 3$ the case $|\iota| \geq p$ was left open in [6] (see [6, Remark 18] for $|\iota| = p$).

Results. We settle the remaining case for Banach spaces whose norms need not be ultrametric, over every base field.

Theorem 1 (Main Theorem). *Let K be a nontrivially normed field of characteristic $p > 0$, and let $k \geq p$. There are Banach spaces E and F over K such that, for every finite set ι with $|\iota| = k$, the map*

$$Q: \mathcal{L}(E, E) \rightarrow \mathcal{L}(\text{Alt}^\iota(E; F), \text{Alt}^\iota(E; F)), \quad Q(f)(m) = m \circ (f, \dots, f),$$

is analytic at no point of $\mathcal{L}(E, E)$. Equivalently, there is no bounded k -linear map P on $\mathcal{L}(E, E)$ with $P(f, \dots, f) = Q(f)$ for all f . The Banach space F admits no equivalent ultrametric norm.

The spaces are explicit (§6). Choose $t \in K$ with $0 < |t| < 1$, let $\widehat{K}$ be the completion of K and $K_1 \subseteq \widehat{K}$ the closure of $\mathbb{F}_p(t)$, which is isometrically isomorphic to the Laurent series field $\mathbb{F}_p((X))$ with $|X| = |t|$. Put $E_1 := \ell^\infty(\mathbb{N}, K_1)$, and let B be the completion of the exterior power $\Lambda_{K_1}^k E_1$ under the projective norm. Then E and F are the completed projective tensor products $E_1 \widehat{\otimes}_\pi \widehat{K}$ and $B \widehat{\otimes}_\pi \widehat{K}$ over K_1 , regarded as K -Banach spaces.

Corollary 2. *Let K be any nontrivially normed field and ι any finite set. Then $Q_{E, E', F}^K$ is analytic for all K -Banach spaces E, E', F if and only if $|\iota|! \neq 0$ in K . More precisely:*

- (1) *if $|\iota|! \neq 0$ in K , then $Q_{E, E', F}^K$ is analytic, indeed given at every point by a finite power series, for all normed spaces E, E', F , complete or not;*
- (2) *if $|\iota|! = 0$ in K , then there are Banach spaces $E = E'$ and F , with F admitting no equivalent ultrametric norm, such that $Q_{E, E, F}^K$ is analytic at no point.*

In Mathlib's terms: for the spaces of Theorem 1, `compContinuousLinearMapCLM` satisfies `ContDiff K n` for every $n : \mathbb{N} \cup \{\infty\}$ by [7] (on the branch of that draft pull request), and `ContDiffAt K ω` fails at every point. So the range of exponents in [7] is *sharp* over every field of characteristic p as soon as $|\iota| \geq p$. By Corollary 2, it cannot be extended to ω for general Banach spaces over any field in which $|\iota|!$ vanishes.

By the third item above (Theorem A, quoted from that unpublished formalization), no counterexample, over any K , can have a spherically complete ultrametric target (a nonzero ultrametric normed space forces K to be ultrametric). Over a discretely valued base, such as $\mathbb{F}_q((t))$, the target of a Banach counterexample cannot even carry an equivalent ultrametric

norm; we deduce this from Theorem A in Proposition 43, so it rests on the same formalization. The target of Theorem 1 has no equivalent ultrametric norm over any base (Proposition 41). For ultrametric targets the question has a different answer, which depends on the base field and is partly open (§7).

Strategy. *Stage 1 (§2).* Analyticity of Q at a single point is equivalent to the existence of a bounded k -linear lift P with $P(f, \dots, f) = Q(f)$: the lift is the k -th Taylor coefficient, identified by uniqueness of power series in one variable. No division by $k!$ occurs.

Stage 2 (§4). Over $K_1 = \kappa((X))$ with κ finite of characteristic p , a bounded lift P for $E_1 = \ell^\infty(\mathbb{N}, K_1)$ and the target B produces, by evaluation at multiplication operators and at the universal alternating map $x \mapsto x_1 \wedge \dots \wedge x_k$, followed by the constant-coefficient map, a κ -multilinear map

$$\Psi: E_0^k \times E_0^k \rightarrow \Lambda_\kappa^k E_0, \quad E_0 = \kappa^\mathbb{N},$$

which is alternating in the last k slots, satisfies the identity

$$\sum_{\sigma \in S_k} \Psi(u_{\sigma(1)}, \dots, u_{\sigma(k)}; x) = \sum_{\sigma \in S_k} (u_{\sigma(1)}x_1) \wedge \dots \wedge (u_{\sigma(k)}x_k)$$

(the multilinear instance of the polarized diagonal identity; Ψ satisfies even the full polarized identity, Lemma 25), and whose values all have support dimension at most a fixed d . The point is that the constant-coefficient map is κ -linear but not K_1 -linear, and has bounded support on bounded subsets of B (Proposition 24).

Stage 3 (§3). Over a finite field with $k! = 0$ no such Ψ exists (Theorem 15). Ramsey's theorem makes the coefficients of Ψ depend only on the order pattern of their indices. Sign-weighted clusters of four consecutive indices then cancel every term that is not attached to the output, a staircase of clusters and the support bound force the cluster value $\chi(\tau)$, $\tau \in S_k$, to be invariant under adjacent transpositions, and the diagonal identity gives $\sum_\tau \chi(\tau) = 1$. Hence $k! \chi = 1$, which is impossible. When $k!$ is invertible the same computation returns $\chi = 1/k!$, the value realized by normalized alternatization.

Stage 4 (§5). Every $\widehat{K}$ of characteristic p contains a closed, spherically complete copy K_1 of $\mathbb{F}_p((X))$. Ingleton's Hahn–Banach theorem [3] gives a norm-one K_1 -linear projection $\widehat{K} \rightarrow K_1$. It makes the projective base change $V \mapsto V \widehat{\otimes}_\pi \widehat{K}$ well behaved for arbitrary, not necessarily ultrametric, K_1 -Banach spaces V , and it lets a bounded lift over $\widehat{K}$ descend to one over K_1 . An incomplete K is reduced to $\widehat{K}$ by density.

Stage 5 (§6). Sums of N unit wedges on disjoint blocks of indices have norm between N/M_r and N in B , hence in F , where $r = |t|$ and $M_r := \max_{l \geq 0} (l+1)r^l$ (so $M_r = 1$ for $r \leq \frac{1}{2}$). So F has no equivalent ultrametric norm.

What is machine-checked. The lift criterion of Stage 1 and the positive direction of Corollary 2 are formalized in Lean 4 with Mathlib. The C^n statement of [7], the quoted Theorem A, a few auxiliary lemmas and the characteristic-2 counterexample of [6] (a different construction) are also formalized (§8). Theorem 1 is proved on paper only; it uses Ramsey's and Ingleton's theorems from the literature and is otherwise self-contained.

Organization. §2 collects conventions, the lift criterion and facts about exterior powers. §3 proves the finite-field theorem. §4 constructs the target over $\kappa((X))$, κ finite, and proves the Main Theorem over that field. §5 carries it to an arbitrary base field. §6 assembles the proofs of Theorem 1 and Corollary 2. §7 discusses ultrametric targets, related results and open problems, and §8 describes the formalization.

2. PRELIMINARIES

2.1. Conventions. We write $\mathbb{N} = \{0, 1, 2, \dots\}$. Throughout, K denotes the base field and k a degree ([6] writes k for the field). A *normed field* K carries a multiplicative absolute value $|\cdot|$ satisfying the triangle inequality. It is *nontrivially normed* if some $c \in K$ has $0 < |c| < 1$, and *ultrametric* if $|x + y| \leq \max(|x|, |y|)$. A *normed K -space* X satisfies $\|\lambda x\| = |\lambda| \|x\|$; it is *Banach* if it is complete. We do *not* require norms of normed spaces to be ultrametric. (In non-archimedean functional analysis “Banach” often includes “ultrametric”; for that reading see §7.) We write $\mathcal{L}(X, Y)$ for the bounded linear maps, $\text{Mult}^k(X; Y)$ for the bounded k -linear maps $X^k \rightarrow Y$, and $\text{Alt}^k(X; Y) \subseteq \text{Mult}^k(X; Y)$ for the closed subspace of alternating ones, all with the operator norm; a subscript K records the field when needed. An alternating map m is antisymmetric: expanding $0 = m(\dots, y + z, \dots, y + z, \dots)$ gives $m(\dots, y, \dots, z, \dots) = -m(\dots, z, \dots, y, \dots)$, and hence $m(x_{\sigma(1)}, \dots, x_{\sigma(k)}) = \text{sgn}(\sigma) m(x_1, \dots, x_k)$ for $\sigma \in S_k$. In characteristic 2 the converse fails.

For a finite set ι with $|\iota| = k$, a bijection $\iota \cong \{1, \dots, k\}$ identifies $\text{Alt}^\iota(E; F)$ isometrically with $\text{Alt}^k(E; F)$ and the map (1.1) with the corresponding map for $\{1, \dots, k\}$; nothing below depends on the choice. So we work with

$$Q = Q_{E, E', F}^K: \mathcal{L}(E, E') \rightarrow \mathcal{L}(\text{Alt}^k(E'; F), \text{Alt}^k(E; F)),$$

where $Q(f)(m)(x_1, \dots, x_k) = m(fx_1, \dots, fx_k)$.

Definition 3. A *bounded k -linear lift* of Q is a bounded k -linear map $P: \mathcal{L}(E, E')^k \rightarrow \mathcal{L}(\text{Alt}^k(E'; F), \text{Alt}^k(E; F))$ with $P(f, \dots, f) = Q(f)$ for all $f \in \mathcal{L}(E, E')$.

A map $g: X \rightarrow Z$ between normed spaces is *analytic at x_0* if there are bounded n -linear maps $p_n: X^n \rightarrow Z$ ($n \geq 0$) and a radius $R > 0$ with $\sup_n \|p_n\| R^n < \infty$ such that $g(x_0 + h) = \sum_{n \geq 0} p_n(h, \dots, h)$, a convergent series, whenever $\|h\| < R$. Up to shrinking R this is Mathlib’s `AnalyticAt` [5], and in Mathlib `ContDiffAt K ω` at a point implies `AnalyticAt` there (`ContDiffAt.analyticAt`).

Lemma 4. A normed field of characteristic $p > 0$ is ultrametric.

Proof. For $n \in \mathbb{N}$ the element $n \cdot 1$ lies in the prime field $\mathbb{F}_p$, so $(n \cdot 1)^p = n \cdot 1$ and $|n \cdot 1| \in \{0, 1\}$. Hence $|(x + y)^N| \leq \sum_{i=0}^N \binom{N}{i} |x|^i |y|^{N-i} \leq (N + 1) \max(|x|, |y|)^N$ for every N . Taking N -th roots and letting $N \rightarrow \infty$ gives $|x + y| \leq \max(|x|, |y|)$. $\square$

2.2. The lift criterion.

Lemma 5 (one-variable uniqueness). *Let K be nontrivially normed and Z a normed K -space, not necessarily complete. Let $z_0, z_1, \dots \in Z$, $\rho > 0$ and $M \geq 0$ with $\|z_n\| \rho^n \leq M$ for all n . If $\sum_n t^n z_n$ converges to 0 for every $t \in K$ with $|t| < \rho$, then $z_n = 0$ for all n .*

Proof. Otherwise let N be least with $z_N \neq 0$. For $0 < |t| < \rho/2$ the series $\sum_{n \geq N} t^{n-N} z_n$ converges, being t^{-N} times the convergent tail $\sum_{n \geq N} t^n z_n$, and $t^N (z_N + \sum_{n \geq N} t^{n-N} z_n) = 0$. Hence

$$\|z_N\| = \left\| \sum_{n \geq N} t^{n-N} z_n \right\| \leq \sum_{n \geq N} M \rho^{-n} |t|^{n-N} = M \rho^{-N} \frac{|t|/\rho}{1 - |t|/\rho},$$

where the norm of a convergent series is bounded by the sum of the norms by continuity of the norm on partial sums. Since K is nontrivially normed, there are $t \neq 0$ with $|t|$ arbitrarily small, so $\|z_N\| = 0$, a contradiction. $\square$

Proposition 6 (lift criterion). *Let K be any nontrivially normed field, E, E', F normed K -spaces and $k \geq 1$.*

- (1) If Q has a bounded k -linear lift P , then at every $f_0 \in \mathcal{L}(E, E')$ the map Q is given by a finite power series of infinite radius; in particular Q is analytic at every point.
- (2) If Q is analytic at some point f_0 , with power series $(p_n)_n$ there, then p_k is a bounded k -linear lift of Q .
- (3) Hence the following are equivalent: Q has a bounded k -linear lift; Q is analytic at every point; Q is analytic at some point. If they fail, then Mathlib's `ContDiffAt` $K \omega Q f_0$ fails at every f_0 .

Proof. (1) For $y \in \mathcal{L}(E, E')$, multilinearity gives

$$Q(f_0 + y) = P(f_0 + y, \dots, f_0 + y) = \sum_{r=0}^k q_r(y, \dots, y),$$

where $q_r(y_1, \dots, y_r)$ is the sum, over the r -element subsets $S \subseteq \{1, \dots, k\}$, of P evaluated with $y_1, \dots, y_r$ in the slots of S in increasing order and f_0 in the other slots. Each q_r is bounded r -linear, with $\|q_r\| \leq \binom{k}{r} \|P\| \|f_0\|^{k-r}$, so $(q_r)_{r \leq k}$ is a power series of infinite radius for Q at f_0 .

(2) Let $\text{incl}: \text{Alt}^k(E; F) \hookrightarrow \text{Mult}^k(E; F)$ be the isometric inclusion, and put

$$Z := \mathcal{L}(\text{Alt}^k(E'; F), \text{Mult}^k(E; F)), \quad \mathcal{I}(T) := \text{incl} \circ T.$$

Then $\mathcal{I}$ is a linear isometry into Z , hence injective. Fix $h \in \mathcal{L}(E, E')$, $h \neq 0$. For $m \in \text{Alt}^k(E'; F)$, $x \in E^k$ and $t \in K$, multilinearity of m gives the *ambient polynomial identity*

$$(2.1) \quad \mathcal{I}(Q(f_0 + th)) = \sum_{r=0}^k t^r A_r(h), \quad A_r(h)(m)(x) := \sum_{|S|=r} m(u_1^S, \dots, u_k^S),$$

where $u_i^S = hx_i$ for $i \in S$ and $u_i^S = f_0x_i$ for $i \notin S$. Here $A_r(h) \in Z$ is bounded, with $\|A_r(h)\| \leq \binom{k}{r} \|h\|^r \|f_0\|^{k-r}$, and $A_k(h) = \mathcal{I}(Q(h))$. (For $r < k$ we do not need to know whether $A_r(h)(m)$ is alternating; this is why (2.1) is written in Z .) Let R and $M' := \sup_n \|p_n\| R^n$ be as in the definition of analyticity, and put $\rho := R/\|h\|$. For $|t| < \rho$ we have $\|th\| < R$, so $Q(f_0 + th) = \sum_n t^n p_n(h, \dots, h)$, and applying $\mathcal{I}$ and (2.1) gives $\sum_n t^n z_n = 0$ in Z , with $z_n := \mathcal{I}(p_n(h, \dots, h)) - A_n(h)$ and $A_n(h) := 0$ for $n > k$. Since $\|z_n\| \rho^n \leq M' + \max_{r \leq k} \|A_r(h)\| \rho^r$, Lemma 5 gives $z_k = 0$, that is, $\mathcal{I}(p_k(h, \dots, h)) = \mathcal{I}(Q(h))$, hence $p_k(h, \dots, h) = Q(h)$. For $h = 0$ both sides vanish since $k \geq 1$.

(3) follows from (1) and (2), and from `ContDiffAt.analyticAt` for the last clause. $\square$

Remark 7. Neither proof divides by $k!$. Item (2) says more than that *some* lift exists: every power series of Q at every point has its degree- k term equal to a lift.

2.3. Exterior powers, supports and determinant arrays. Let L be a field, V an L -vector space and $k \geq 1$. We write $\Lambda^k V = \Lambda_L^k V$ for the algebraic exterior power, in which $x_1 \wedge \dots \wedge x_k = 0$ whenever two x_i coincide. We use the standard facts: for a subspace $W \subseteq V$ the map $\Lambda^k W \rightarrow \Lambda^k V$ is injective, and we regard $\Lambda^k W$ as a subspace of $\Lambda^k V$; if $u_1, \dots, u_d$ is a basis of a finite-dimensional space U , the wedges $u_I := u_{i_1} \wedge \dots \wedge u_{i_k}$, $I = \{i_1 < \dots < i_k\}$, form a basis of $\Lambda^k U$; and every $\omega \in \Lambda^k V$ lies in $\Lambda^k U$ for some finite-dimensional $U \subseteq V$.

Definition 8. The *support dimension* of $\omega \in \Lambda^k V$ is

$$\text{sdim}(\omega) := \min\{\dim W : W \subseteq V \text{ finite-dimensional, } \omega \in \Lambda^k W\}.$$

Clearly $\text{sdim}(0) = 0$, $\text{sdim}(c\omega) = \text{sdim}(\omega)$ for $c \neq 0$, $\text{sdim}(\omega + \omega') \leq \text{sdim}(\omega) + \text{sdim}(\omega')$ (use $W + W'$), and $\text{sdim}(\omega) \leq \dim \text{span } S$ if ω is a sum of wedges of vectors from a finite set S . If $V \subseteq V'$, then sdim computed in V' is at most sdim computed in V .

Let S be a set and $V \subseteq L^S$ a space of functions on S . The *determinant array* is the linear map

$$\Omega: \Lambda^k V \rightarrow L^{S^k}, \quad \Omega_{y_1 \wedge \dots \wedge y_k}(c_1, \dots, c_k) := \det(y_b(c_a))_{a,b=1}^k.$$

It is well defined because the right side is multilinear and alternating in $(y_1, \dots, y_k)$ (these are the columns). For each ω , the function Ω_ω is alternating in $(c_1, \dots, c_k)$ (the rows); in particular $\Omega_\omega(c_{\sigma(1)}, \dots, c_{\sigma(k)}) = \text{sgn}(\sigma) \Omega_\omega(c_1, \dots, c_k)$.

Lemma 9. *The determinant array $\Omega: \Lambda^k V \rightarrow L^{S^k}$ is injective.*

Proof. Let $\Omega_\omega = 0$ and choose a finite-dimensional $U \subseteq V$ with $\omega \in \Lambda^k U$. The evaluations $\text{ev}_s|_U$, $s \in S$, separate the points of U (a function in U vanishing at every s is 0), so they span the dual space U^* . Choose $s_1, \dots, s_d \in S$ such that the $\text{ev}_{s_i}|_U$ form a basis of U^* , and let $u_1, \dots, u_d$ be the dual basis of U , so that $u_j(s_i) = \delta_{ij}$. For increasing k -subsets I, J of $\{1, \dots, d\}$ we get $\Omega_{u_I}(s_{j_1}, \dots, s_{j_k}) = \det(u_{i_b}(s_{j_a}))_{a,b} = \delta_{IJ}$. Writing $\omega = \sum_I c_I u_I$ therefore gives $0 = \Omega_\omega(s_{j_1}, \dots, s_{j_k}) = c_J$ for every J , so $\omega = 0$. $\square$

Lemma 10 (flattening rank is at most the support). *Let $V \subseteq L^S$, $\omega \in \Lambda^k V$ and $\alpha \in \{1, \dots, k\}$. Let $R \subseteq S$ and $R' \subseteq S^{\{1, \dots, k\} \setminus \{\alpha\}}$ be finite. The $R \times R'$ matrix M with entries $M(c_\alpha, c') := \Omega_\omega(c)$, where c has c_α in slot α and the entries of c' in the other slots, has rank at most $\text{sdim}(\omega)$.*

Proof. Write $\omega = \sum_j y_{j,1} \wedge \dots \wedge y_{j,k}$ with all $y_{j,b}$ in a subspace W of dimension $\text{sdim}(\omega)$. Expanding each determinant along row α , $\det(y_{j,b}(c_a))_{a,b} = \sum_b \pm y_{j,b}(c_\alpha) \cdot \mu_{j,b}(c')$, where the minor $\mu_{j,b}(c')$ does not involve c_α . So for fixed c' the column $c_\alpha \mapsto M(c_\alpha, c')$ is a linear combination of the restrictions $y_{j,b}|_R$. All columns lie in the restriction of W to R , a space of dimension at most $\text{sdim}(\omega)$. $\square$

Lemma 11 (contraction). *Let $k \geq 2$ and $\varphi = (\varphi_1, \dots, \varphi_{k-1}) \in (V^*)^{k-1}$. There is a linear map $c_\varphi: \Lambda^k V \rightarrow V$ with*

$$c_\varphi(x_1 \wedge \dots \wedge x_k) = \sum_{i=1}^k (-1)^{k+i} \det(\varphi_a(x_b))_{1 \leq a \leq k-1, b \neq i} x_i,$$

and $c_\varphi(\Lambda^k W) \subseteq W$ for every subspace W . Consequently $\text{sdim}(\omega) \geq \dim \text{span}\{c_\varphi(\omega) : \varphi \in (V^)^{k-1}\}$.*

Proof. The right side is multilinear in (x_b) . For $\psi \in V^*$, applying ψ to it gives, by Laplace expansion along the last row, the determinant of the $k \times k$ matrix with entries $\varphi_a(x_b)$ in the rows $a < k$ and $\psi(x_b)$ in the last row; it vanishes when two columns coincide. Since V^* separates the points of V , the right side vanishes whenever $x_a = x_b$ for some $a \neq b$, so it factors through $\Lambda^k V$. Each term is a multiple of some x_i , which gives $c_\varphi(\Lambda^k W) \subseteq W$. If $\omega \in \Lambda^k W$ with $\dim W = \text{sdim}(\omega)$, all $c_\varphi(\omega)$ lie in W . $\square$

3. THE MULTIPLIER THEOREM OVER A FINITE FIELD

3.1. Lifts and three diagonal identities. Fix a field L , an integer $k \geq 1$, L -vector spaces A and V , and a linear map $D: A \rightarrow \text{End}_L(V)$, $a \mapsto D_a$. A *lift of D in degree k* is a $2k$ -linear map $\Psi: A^k \times V^k \rightarrow \Lambda^k V$ over L that is alternating in the V -slots. We write $[k] := \{1, \dots, k\}$. For $m \geq 1$, $f: [k] \rightarrow [m]$, $b = (b_1, \dots, b_m) \in A^m$ and $\lambda \in L^m$ put

$$b_f := (b_{f(1)}, \dots, b_{f(k)}), \quad \text{type } f := (|f^{-1}(1)|, \dots, |f^{-1}(m)|) \in \mathbb{N}^m, \quad \lambda_f := \prod_{j=1}^k \lambda_{f(j)} = \lambda^{\text{type } f}.$$

Three diagonal identities will be compared. The *pointwise identity* is

$$(Pw) \quad \Psi(a, \dots, a; x_1, \dots, x_k) = D_a x_1 \wedge \dots \wedge D_a x_k \quad \text{for all } a \in A, x \in V^k.$$

The *polarized identity* is: for all $m \geq 1$, $b \in A^m$, $\alpha \in \mathbb{N}^m$ with $|\alpha| = k$, and $x \in V^k$,

$$(Pol) \quad \sum_{\text{type } f=\alpha} \Psi(b_f; x) = \sum_{\text{type } f=\alpha} D_{b_{f(1)}} x_1 \wedge \dots \wedge D_{b_{f(k)}} x_k.$$

Its *multilinear instance* is: for all $b_1, \dots, b_k \in A$ and $x \in V^k$,

$$(Poll) \quad \sum_{\sigma \in S_k} \Psi(b_{\sigma(1)}, \dots, b_{\sigma(k)}; x) = \sum_{\sigma \in S_k} D_{b_{\sigma(1)}} x_1 \wedge \dots \wedge D_{b_{\sigma(k)}} x_k.$$

Here $\sum_{\text{type } f=\alpha}$ runs over the maps $f: [k] \rightarrow [m]$ of type α . (Poll) is the instance $m = k$, $\alpha = (1, \dots, 1)$ of (Pol), since the maps of type $(1, \dots, 1)$ are the permutations. For $b \in A^m$ and $x \in V^k$ put $a_\lambda := \sum_i \lambda_i b_i$ and

$$\begin{aligned} \Delta_{b,x}(\lambda) &:= \Psi(a_\lambda, \dots, a_\lambda; x) - D_{a_\lambda} x_1 \wedge \dots \wedge D_{a_\lambda} x_k = \sum_{|\alpha|=k} \lambda^\alpha \delta_\alpha(b, x), \\ \delta_\alpha(b, x) &:= \sum_{\text{type } f=\alpha} \left[\Psi(b_f; x) - D_{b_{f(1)}} x_1 \wedge \dots \wedge D_{b_{f(k)}} x_k \right]. \end{aligned}$$

The expansion uses multilinearity of Ψ in the A -slots, linearity of D and multilinearity of the wedge. So $\Delta_{b,x}$ is a homogeneous polynomial of degree k in $\lambda = (\lambda_1, \dots, \lambda_m)$ with coefficients in $\Lambda^k V$: (Pol) says that every $\Delta_{b,x}$ is the zero polynomial, and (Pw) says that every $\Delta_{b,x}$ vanishes at every point of L^m (for the converse take $m = 1$ and $\lambda = 1$).

Lemma 12 (identity principle). *Let $\mathbb{K}$ be an infinite field, Y a $\mathbb{K}$ -vector space and $g(\lambda) = \sum_\alpha \lambda^\alpha y_\alpha$ a polynomial in $\lambda \in \mathbb{K}^m$ with finitely many nonzero coefficients $y_\alpha \in Y$. If $g(\lambda) = 0$ for all $\lambda \in \mathbb{K}^m$, then every $y_\alpha = 0$.*

Proof. The y_α span a finite-dimensional subspace. Its coordinate functionals with respect to a basis turn g into finitely many scalar polynomials vanishing on $\mathbb{K}^m$. These are zero, by induction on m , since a nonzero polynomial in one variable has finitely many roots. $\square$

Lemma 13. *Let q be a prime power, $L' \supseteq \mathbb{F}_q$ a field, $1 \leq e \leq q$, and $g \in L'[\lambda_1, \dots, \lambda_m]$ homogeneous of degree e with $g(\lambda) = 0$ for every $\lambda \in \mathbb{F}_q^m$. Then $g = 0$.*

Proof. Call a monomial λ^β *reduced* if every $\beta_i \leq q-1$. The indicator function of a point $a \in \mathbb{F}_q^m$ is $\prod_i (1 - (\lambda_i - a_i)^{q-1})$, which expands into reduced monomials. So the q^m reduced monomials span the q^m -dimensional L' -space of functions $\mathbb{F}_q^m \rightarrow L'$, and they are linearly independent as functions. A monomial of degree $e \leq q$ that is not reduced is λ_i^q (and then $e = q \geq 2$); as a function on $\mathbb{F}_q^m$ it agrees with the reduced monomial λ_i , of degree $1 < e$. Replacing each λ_i^q in g by λ_i therefore sends the monomials of g injectively to reduced monomials and does not change g as a function. The resulting reduced polynomial vanishes as a function, hence is 0, and so $g = 0$. $\square$

Proposition 14 (pointwise versus polarized). *Let Ψ be a lift of D in degree k over a field L .*

- (1) (Pol) $\Rightarrow$ (Pw) $\Rightarrow$ (Poll), over every field.
- (2) If L is infinite, or if $L = \mathbb{F}_q$ and $k \leq q$, then (Pw) $\Rightarrow$ (Pol).
- (3) If $L = \mathbb{F}_q$ and $k \geq q + 1$, then (Pw) does not imply (Pol) in general.
- (4) If $k! = 0$ in L , then (Poll) does not imply (Pw) in general.

Proof. (1) $(\text{Pol}) \Rightarrow (\text{Pw})$ is the case $m = 1$, $\lambda = 1$. For $(\text{Pw}) \Rightarrow (\text{Poll})$, let $b_1, \dots, b_k \in A$, $x \in V^k$, and put $b_S := \sum_{i \in S} b_i$ for $S \subseteq [k]$. Then

$$\begin{aligned} \sum_{S \subseteq [k]} (-1)^{k-|S|} \Psi(b_S, \dots, b_S; x) &= \sum_S (-1)^{k-|S|} \sum_{f: [k] \rightarrow S} \Psi(b_f; x) \\ &= \sum_{f: [k] \rightarrow [k]} \Psi(b_f; x) \sum_{S \supseteq f([k])} (-1)^{k-|S|} = \sum_{f \text{ bijective}} \Psi(b_f; x), \end{aligned}$$

because $\sum_{S \supseteq T} (-1)^{k-|S|} = (1-1)^{k-|T|}$ is 1 if $T = [k]$ and 0 otherwise. The same computation applies to the multilinear map $(a_1, \dots, a_k) \mapsto D_{a_1} x_1 \wedge \dots \wedge D_{a_k} x_k$, whose diagonal is the right side of (Pw) . So (Pw) at the 2^k points b_S gives (Poll) . All coefficients are integers.

(2) The coefficients of $\Delta_{b,x}$ span a finite-dimensional subspace of $\Lambda^k V$; in a basis of it each coordinate of $\Delta_{b,x}$ is a scalar form of degree k vanishing on L^m . It is zero by Lemma 12 if L is infinite, and by Lemma 13 if $L = \mathbb{F}_q$ and $k \leq q$.

(3) Let $\dim A \geq 3$, with $b_1, b_2, b_3 \in A$ and functionals g_1, g_2, g_3 on A such that $g_i(b_j) = \delta_{ij}$, and let $y_1, \dots, y_k \in V$ be linearly independent. Put

$$\theta(a_1, \dots, a_k) := [g_1(a_1) \cdots g_1(a_q) g_2(a_{q+1}) - g_1(a_1) g_2(a_2) \cdots g_2(a_{q+1})] \cdot g_3(a_{q+2}) \cdots g_3(a_k)$$

(for $k = q+1$ the g_3 -factor is empty and $\dim A \geq 2$ suffices), and $\Theta(a; x) := \theta(a) x_1 \wedge \dots \wedge x_k$. Then Θ is a lift of $D = 0$. It satisfies (Pw) , because $\theta(a, \dots, a) = (g_1(a)^q g_2(a) - g_1(a) g_2(a)^q) g_3(a)^{k-q-1} = 0$ since $c^q = c$ for $c \in \mathbb{F}_q$. It violates (Pol) for $\alpha = (q, 1, k-q-1)$ and $b = (b_1, b_2, b_3)$: the first product contributes to $\sum_{\text{type } f = \alpha} \theta(b_f)$ exactly for $f = (1, \dots, 1, 2, 3, \dots, 3)$, with value 1; the second product would need $f(2) = \dots = f(q+1) = 2$, that is, at least $q \geq 2$ values equal to 2, which is impossible for type α . So the coefficient $\delta_\alpha(b, y)$ of Θ is $y_1 \wedge \dots \wedge y_k \neq 0$.

(4) Take $\bar{a} \in A$ and a functional g_1 with $g_1(\bar{a}) = 1$, and $y_1, \dots, y_k$ linearly independent. The lift $\Theta'(a; x) := g_1(a_1) \cdots g_1(a_k) x_1 \wedge \dots \wedge x_k$ of $D = 0$ satisfies (Poll) , since $\sum_\sigma \Theta'(b_\sigma; x) = k! g_1(b_1) \cdots g_1(b_k) x_1 \wedge \dots \wedge x_k = 0$ for all $b_1, \dots, b_k \in A$ and $x \in V^k$, but $\Theta'(\bar{a}, \dots, \bar{a}; y) = y_1 \wedge \dots \wedge y_k \neq 0$. $\square$

So over $\mathbb{F}_q$ the pointwise identity at $\mathbb{F}_q$ -points is strictly weaker than the polarized one as soon as $k > q$. The proof below uses only (Poll) , which by Proposition 14(1) is implied by either. The lift built in §4 satisfies the full (Pol) in every degree, because it comes from polarizing over an *infinite* field (Lemma 25).

3.2. The theorem. From now on in this section L is a *finite* field with $k! = 0$ in L , that is, $\text{char } L = p \leq k$; in particular $k \geq 2$. Put $V := L^{\mathbb{N}}$ and let $V_{\text{fin}} := L^{(\mathbb{N})} \subseteq V$ be the subspace of finitely supported sequences. For $u, x \in V$ we write ux for the coordinatewise product; so $D_u x = ux$ is the multiplier family.

Theorem 15 (multipliers over a finite field). *Let L be a finite field and $k \geq 1$ with $k! = 0$ in L .*

- (1) *There is no $2k$ -linear map $\Psi: V_{\text{fin}}^k \times V_{\text{fin}}^k \rightarrow \Lambda_L^k V$ over L such that*
 - (a) *Ψ is antisymmetric in the last k slots;*
 - (b) *Ψ satisfies (Poll) for $D_u x = ux$: for all $u_1, \dots, u_k, x_1, \dots, x_k \in V_{\text{fin}}$,*

$$\sum_{\sigma \in S_k} \Psi(u_{\sigma(1)}, \dots, u_{\sigma(k)}; x_1, \dots, x_k) = \sum_{\sigma \in S_k} (u_{\sigma(1)} x_1) \wedge \dots \wedge (u_{\sigma(k)} x_k);$$

- (c) *for some $d \in \mathbb{N}$, $\text{sdim}(\Psi(u; x)) \leq d$ for all $u, x \in V_{\text{fin}}^k$ with entries in $\{0, \pm 1\}$.*

- (2) Let $E_0 = V$ or $E_0 = V_{\text{fin}}$. There is no $2k$ -linear map $\Psi: E_0^k \times E_0^k \rightarrow \Lambda_L^k E_0$ over L that is antisymmetric in the last k slots, satisfies (Pol1) for all finitely supported u, x , and has $\text{sdim}(\Psi(u; x)) \leq d$ for all $u, x \in E_0^k$, for some d .

(1) implies (2). Given Ψ as in (2), restrict it to $V_{\text{fin}}^k \times V_{\text{fin}}^k$ and compose with the injective map $\Lambda^k E_0 \rightarrow \Lambda^k V$. Antisymmetry and (Pol1) are preserved, and the support dimension computed in V is at most the one computed in E_0 (§2.3). So the restriction satisfies (a)–(c).

The rest of this section proves (1). Suppose Ψ is as in (1). Write $e_i \in V_{\text{fin}}$ for the unit vectors, $e_a := (e_{a_1}, \dots, e_{a_k})$ for $a \in \mathbb{N}^k$, and Ω for the determinant array of §2.3 with $S = \mathbb{N}$.

3.3. Order-homogeneity from Ramsey's theorem. We use Ramsey's theorem [8] in its infinite form: *for every $n \geq 1$ and every coloring of the n -element subsets of $\mathbb{N}$ with finitely many colors there is an infinite $H \subseteq \mathbb{N}$ all of whose n -element subsets have the same color.*

The pattern of $z \in \mathbb{N}^N$ is the map sending i to the rank of z_i in the set $\{z_1, \dots, z_N\}$. Thus z and z' have the same pattern if and only if $z_i < z_j \Leftrightarrow z'_i < z'_j$ and $z_i = z_j \Leftrightarrow z'_i = z'_j$ for all i, j .

Lemma 16 (pattern homogeneity). *For every $N \geq 1$ and every function $T: \mathbb{N}^N \rightarrow L$ there is an infinite $H \subseteq \mathbb{N}$ such that $T(z) = T(z')$ whenever $z, z' \in H^N$ have the same pattern.*

Proof. Color an N -element set $s = \{s_1 < \dots < s_N\} \subseteq \mathbb{N}$ by the vector

$$(T(s_{\beta(1)}, \dots, s_{\beta(N)}))_{\beta: [N] \rightarrow [N]} \in L^{N^N}.$$

There are finitely many colors because L is finite; this is the only use of the finiteness of L . Ramsey's theorem gives an infinite H on which the color is constant, say equal to $(t_\beta)_\beta$. Let $z \in H^N$ have value set $\{y_1 < \dots < y_r\}$. Pad it with $N - r$ elements of H above y_r to an N -element set $s \subseteq H$; then $s_j = y_j$ for $j \leq r$, and $z = (s_{\beta(1)}, \dots, s_{\beta(N)})$, where $\beta(i)$ is the rank of z_i in $\{y_1, \dots, y_r\}$. The map β depends only on the pattern of z , and $T(z) = t_\beta$. $\square$

Coefficients. For $(a; y; c) \in \mathbb{N}^k \times \mathbb{N}^k \times \mathbb{N}^k$ put

$$T(a; y; c) := \Omega_{\Psi(e_a; e_y)}(c_1, \dots, c_k) \in L.$$

Lemma 16 with $N = 3k$ gives an infinite set $H = \{h_0 < h_1 < h_2 < \dots\}$ on which T depends only on the pattern. For $u_j, v_j \in V_{\text{fin}}$ supported in H , multilinearity of Ψ and linearity of Ω give the finite expansion

$$(3.1) \quad \Omega_{\Psi(u; v)}(c) = \sum_{a, y \in H^k} \prod_{j=1}^k u_j(a_j) \prod_{j=1}^k v_j(y_j) \cdot T(a; y; c).$$

Below, every output index c lies in H^k .

3.4. Clusters and the cluster value. A *cluster* is a set $C = \{h_n, h_{n+1}, h_{n+2}, h_{n+3}\}$ of four consecutive elements of H ; we write $C(i) := h_{n+i-1}$ for $i = 1, \dots, 4$. Since a cluster is an interval of H , two disjoint clusters are *comparable*: every element of one is smaller than every element of the other; we write $C < C'$. Clusters need not be adjacent in H . Fix the weights

$$\begin{aligned} s &= (s_1, s_2, s_3, s_4) := (1, 0, -1, 0) && \text{on the operator slots,} \\ w &= (w_1, w_2, w_3, w_4) := (1, -1, 1, -1) && \text{on the vector slots.} \end{aligned}$$

Over $\mathbb{Z}$, hence over every field,

$$(W) \quad \sum_i s_i = \sum_i w_i = \sum_i s_i w_i = \sum_{i < i'} s_i w_{i'} = \sum_{i > i'} s_i w_{i'} = 0, \quad s_1 w_1 = 1.$$

Indeed $\sum_i s_i = 1 - 1$, $\sum_i w_i = 1 - 1 + 1 - 1$, $s_1 w_1 = 1$, and

$$\begin{aligned} \sum_i s_i w_i &= s_1 w_1 + s_3 w_3 = 1 - 1 = 0, \\ \sum_{i < i'} s_i w_{i'} &= s_1(w_2 + w_3 + w_4) + s_3 w_4 = -1 + 1 = 0, \\ \sum_{i > i'} s_i w_{i'} &= s_3(w_1 + w_2) = -(1 - 1) = 0. \end{aligned}$$

For a cluster C put

$$s_C := \sum_{i=1}^4 s_i e_{C(i)}, \quad w_C := \sum_{i=1}^4 w_i e_{C(i)}.$$

These are vectors of V_{fin} with entries in $\{0, \pm 1\}$, and $s_C w_C = e_{C(1)} - e_{C(3)}$.

The cluster value. For pairwise disjoint clusters $C_1, \dots, C_k$, one for each slot, define

$$\begin{aligned} \chi(C_1, \dots, C_k) &:= \Omega_{\Psi(s_{C_1}, \dots, s_{C_k}; w_{C_1}, \dots, w_{C_k})}(C_1(1), \dots, C_k(1)) \\ &= \sum_{i, i' \in [4]^k} \prod_{j=1}^k s_{i_j} w_{i'_j} \cdot T(C_1(i_1), \dots, C_k(i_k); C_1(i'_1), \dots, C_k(i'_k); C_1(1), \dots, C_k(1)). \end{aligned}$$

Let $\tau \in S_k$ be the *order pattern* of $(C_1, \dots, C_k)$: $\tau(j)$ is the position of C_j among $C_1, \dots, C_k$. The pattern of every $3k$ -tuple in the sum is determined by (i, i') and τ : entries in different clusters compare as the clusters do, and the entries $C_j(i_j)$, $C_j(i'_j)$, $C_j(1)$ in the same cluster compare as their indices do, including equalities. By Lemma 16, $\chi(C_1, \dots, C_k) =: \chi(\tau)$ depends only on τ .

Lemma 17 (cancellation of output-free clusters). *Let $\mathcal{C}_1, \dots, \mathcal{C}_k$ be finite families of clusters that are pairwise disjoint as sets of clusters, such that all clusters in $\bigcup_j \mathcal{C}_j$ are pairwise disjoint. Put $u_j := \sum_{C \in \mathcal{C}_j} s_C$ and $v_j := \sum_{C \in \mathcal{C}_j} w_C$. Let $C_l \in \mathcal{C}_l$ for $l = 1, \dots, k$ and $c := (C_1(1), \dots, C_k(1))$. Then $\Omega_{\Psi(u; v)}(c) = \chi(C_1, \dots, C_k)$.*

Proof. Expand by (3.1). Since the clusters are disjoint, a nonzero term has, for each slot j , an operator index $a_j = C_j^{\text{op}}(i_j)$ and a vector index $y_j = C_j^{\text{vec}}(i'_j)$ with $C_j^{\text{op}}, C_j^{\text{vec}} \in \mathcal{C}_j$, and weight $\prod_j s_{i_j} w_{i'_j}$. Group the terms according to the *cluster assignment* $(C_j^{\text{op}}, C_j^{\text{vec}})_j$.

Call a cluster *free* in a group if it is occupied (it equals some C_j^{op} or some C_j^{vec}) but is none of the output clusters $C_1, \dots, C_k$. Suppose a group has a free cluster Γ . Then $\Gamma \in \mathcal{C}_j$ for exactly one j , because the families are disjoint, so only a_j and/or y_j lie in Γ ; no output index lies in Γ , because $\Gamma \neq C_l$ and the clusters are disjoint. Fix all indices outside Γ and sum over the positions inside Γ . Since Γ is an interval of H containing no other entry of the tuple, every other entry is below all of Γ or above all of Γ , and the pattern of the tuple depends on the positions inside Γ only as follows:

- (i) if only $a_j \in \Gamma$, not at all, and the partial sum is $(\sum_i s_i) \cdot T = 0$;
- (ii) if only $y_j \in \Gamma$, not at all, and the partial sum is $(\sum_i w_i) \cdot T = 0$;
- (iii) if both lie in Γ , only through whether $i_j < i'_j$, $i_j = i'_j$ or $i_j > i'_j$; with the corresponding values $T_<, T_=: T_>$ of T the partial sum is

$$T_< \sum_{i < i'} s_i w_{i'} + T_=: \sum_i s_i w_i + T_> \sum_{i > i'} s_i w_{i'} = 0.$$

(The weights of the other slots are a common factor.) All three vanish by (W), so every group with a free cluster sums to 0. In a remaining group every occupied cluster is an output cluster. The only output cluster in $\mathcal{C}_j$ is C_j , so $C_j^{\text{op}} = C_j^{\text{vec}} = C_j$ for all j . This single group is, by definition, $\chi(C_1, \dots, C_k)$. $\square$

3.5. The staircase and the diagonal.

Lemma 18 (staircase). *Let $\tau \in S_k$, let $\alpha \neq \beta$ be slots with $\tau(\alpha) = \ell$ and $\tau(\beta) = \ell + 1$, and let τ' be τ with these two positions exchanged, so $\tau'(\alpha) = \ell + 1$, $\tau'(\beta) = \ell$ and $\tau'(\gamma) = \tau(\gamma)$ otherwise. Then $\chi(\tau) = \chi(\tau')$.*

Proof. Let $m \geq 1$. Choose pairwise disjoint clusters, in increasing order in H : first a cluster C_γ for each slot γ with $\tau(\gamma) < \ell$, ordered as τ prescribes; then $2m$ clusters $A_1 < B_1 < A_2 < B_2 < \dots < A_m < B_m$; then a cluster C_γ for each slot γ with $\tau(\gamma) > \ell + 1$, ordered as τ prescribes. Put $\mathcal{C}_\alpha := \{A_1, \dots, A_m\}$, $\mathcal{C}_\beta := \{B_1, \dots, B_m\}$ and $\mathcal{C}_\gamma := \{C_\gamma\}$ for the other slots, and let u, v be as in Lemma 17 and $\omega := \Psi(u; v)$. The entries of u and v lie in $\{0, \pm 1\}$, so $\text{sdim}(\omega) \leq d$ by hypothesis (c) of Theorem 15(1).

Consider the $m \times m$ matrix M with rows indexed by $c_\alpha = A_n(1)$ and columns by $c_\beta = B_{n'}(1)$ ($n, n' = 1, \dots, m$), all other output indices being fixed at $c_\gamma = C_\gamma(1)$, and entries $M_{nn'} := \Omega_\omega(c)$. By Lemma 17, applied with the output clusters $A_n \in \mathcal{C}_\alpha$, $B_{n'} \in \mathcal{C}_\beta$ and $C_\gamma \in \mathcal{C}_\gamma$, $M_{nn'}$ is χ of the order pattern of $(A_n, B_{n'}, (C_\gamma)_\gamma)$. Now $A_n < B_{n'}$ if and only if $n \leq n'$; in that case the pattern is τ , and otherwise it is τ' . Hence

$$M = \begin{pmatrix} \chi(\tau) & \chi(\tau) & \cdots & \chi(\tau) \\ \chi(\tau') & \chi(\tau) & \cdots & \chi(\tau) \\ \vdots & \ddots & \ddots & \vdots \\ \chi(\tau') & \cdots & \chi(\tau') & \chi(\tau) \end{pmatrix} = \chi(\tau') \mathbf{1}\mathbf{1}^\top + \delta U,$$

with $\chi(\tau)$ on and above the diagonal and $\chi(\tau')$ below it, where $\delta := \chi(\tau) - \chi(\tau')$, $\mathbf{1} \in L^m$ is the all-ones column, so that $\mathbf{1}\mathbf{1}^\top$ is the all-ones matrix, and $U_{nn'} = [n \leq n']$. M is a submatrix of the flattening of Ω_ω along slot α , so $\text{rank } M \leq \text{sdim}(\omega) \leq d$ by Lemma 10. If $\delta \neq 0$, then δU is invertible (U is upper unitriangular) and $\text{rank } \mathbf{1}\mathbf{1}^\top \leq 1$, so $\text{rank } M \geq m - 1$. Taking $m := d + 2$ gives a contradiction, so $\delta = 0$. $\square$

Lemma 19 (the diagonal). $\sum_{\tau \in S_k} \chi(\tau) = 1$.

Proof. Take clusters $C_1 < \dots < C_k$, and put $u_j := s_{C_j}$, $v_j := w_{C_j}$ and $c := (C_1(1), \dots, C_k(1))$. Apply $\omega \mapsto \Omega_\omega(c)$ to both sides of (Pol1), with the operator vectors $u_1, \dots, u_k$ and the vector arguments $v_1, \dots, v_k$.

Right side. For $i \neq l$ the vectors u_i and v_l have disjoint supports, so $u_i v_l = 0$ and only $\sigma = \text{id}$ survives. It gives $\Omega_{(u_1 v_1) \wedge \dots \wedge (u_k v_k)}(c) = \det((u_b v_b)(C_a(1)))_{a,b}$. Since $u_b v_b = e_{C_b(1)} - e_{C_b(3)}$ and $C_a(1) \neq C_b(3)$ for all a, b , this is the determinant of the identity matrix, so the right side is 1.

Left side, one term. Fix $\sigma \in S_k$ and write $u_\sigma := (u_{\sigma(1)}, \dots, u_{\sigma(k)})$ and $v_\sigma := (v_{\sigma(1)}, \dots, v_{\sigma(k)})$. Antisymmetry of Ψ in the vector slots gives $\Psi(u_\sigma; v) = \text{sgn}(\sigma) \Psi(u_\sigma; v_\sigma)$. For every $\omega \in \Lambda^k V$ the function Ω_ω is alternating in its arguments, so $\Omega_\omega(c) = \text{sgn}(\sigma) \Omega_\omega(c_{\sigma(1)}, \dots, c_{\sigma(k)})$; we apply this with $\omega = \Psi(u_\sigma; v_\sigma)$. Since $\text{sgn}(\sigma)^2 = 1$,

$$\begin{aligned} \Omega_{\Psi(u_\sigma; v)}(c) &= \text{sgn}(\sigma) \Omega_{\Psi(u_\sigma; v_\sigma)}(c) = \text{sgn}(\sigma)^2 \Omega_{\Psi(u_\sigma; v_\sigma)}(c_{\sigma(1)}, \dots, c_{\sigma(k)}) \\ &= \Omega_{\Psi(u_\sigma; v_\sigma)}(c_{\sigma(1)}, \dots, c_{\sigma(k)}). \end{aligned}$$

In the last expression slot j has operator $s_{C_{\sigma(j)}}$, vector $w_{C_{\sigma(j)}}$ and output $C_{\sigma(j)}(1)$, so it equals $\chi(C_{\sigma(1)}, \dots, C_{\sigma(k)})$ by definition. Its order pattern is σ , since $C_{\sigma(j)}$ is the $\sigma(j)$ -th cluster. (Lemma 17 is not needed here.)

Left side, summed. As σ runs over S_k , so does the order pattern, once each. Comparing with the right side,

$$\sum_{\sigma \in S_k} \Omega_{\Psi(u_\sigma; v)}(c) = \sum_{\tau \in S_k} \chi(\tau) = 1. \quad \square$$

Proof of Theorem 15. The position transpositions $\tau \mapsto (\ell \ \ell+1) \circ \tau$, $1 \leq \ell < k$, generate the action of S_k on itself by left multiplication, and Lemma 18 says that χ is invariant under each of them. So $\chi \equiv \chi_0$ is constant, and Lemma 19 gives $k! \chi_0 = 1$ in L . Since $k! = 0$ in L , this reads $0 = 1$. This proves (1), and (2) follows as shown above. $\square$

Remark 20 (where the hypotheses enter). The proof uses only multilinearity, antisymmetry in the vector slots, the support bound at the finitely many test inputs of Lemma 18 (Lemma 19 does not use the bound), and (Pol1) at the test inputs of Lemma 19. The finiteness of L enters only in Lemma 16, and $k! = 0$ only in the last line. By the inclusion–exclusion in the proof of Proposition 14(1), which uses only multilinearity in the operator slots, the theorem remains true with the pointwise identity (Pw) at L -points in place of (Pol1), for this family of multipliers. In characteristic 2 antisymmetry is the same as symmetry, and the two signs in Lemma 19 are trivial.

Remark 21 (when $k!$ is invertible). Nothing goes wrong if $k! \neq 0$ in L . Then the *normalized alternatization*

$$\Psi_{\text{alt}}(u; x) := \frac{1}{k!} \sum_{\sigma \in S_k} (u_{\sigma(1)}x_1) \wedge \cdots \wedge (u_{\sigma(k)}x_k)$$

is alternating in x , satisfies (Pol1), and $\text{sdim}(\Psi_{\text{alt}}(u; x)) \leq k^2$. Lemmas 16–19 hold for it, and the computation above returns $\chi \equiv 1/k!$: for pairwise disjoint clusters only $\sigma = \text{id}$ survives in $\Psi_{\text{alt}}(s_{C_1}, \dots, s_{C_k}; w_{C_1}, \dots, w_{C_k})$, and the determinant at $(C_1(1), \dots, C_k(1))$ is that of the identity. The final equation $k! \cdot (1/k!) = 1$ is then consistent. So the obstruction is exactly the vanishing of $k!$, not positive characteristic as such. For example, at $(p, k) = (5, 3)$ every step holds and, since $3! = 6 = 1$ in $\mathbb{F}_5$, the chain stops at $\chi_0 = 1$.

Remark 22 (another possible approach). We sketch an alternative way to obtain the input needed in §4. Nothing in this paper depends on it.

The unproved input. The sketch rests on a statement that we do *not* prove here: the analogue of Theorem 15 over infinite fields L with $k! = 0$ in L , for lifts that satisfy (Pw) and have uniformly bounded support, of an arbitrary linear family $D: A \rightarrow \text{End}_L(V)$ whose image modulo the finite-rank operators is infinite-dimensional.

Base change and minors. Let Ψ be a lift over $\mathbb{F}_q$ whose values have support dimension at most d . After the finite base change $\mathbb{F}_q \subseteq \mathbb{F}_{q^N}$, the values of the base change of Ψ at $\mathbb{F}_{q^N}$ -points have support dimension at most $d_N := N^{2k}d$, since each input has only N pieces over $\mathbb{F}_q$; choose N with $q^N > d_N + 1$. One would then use that the support dimension of $\omega \in \Lambda^k V$ equals the rank of its contraction matrix, the matrix of coordinates of the vectors $c_\varphi(\omega)$ of Lemma 11, with φ running over tuples of dual basis vectors. Lemma 11 gives only the inequality $\geq$, and we do not prove the equality. Granting it, the vanishing of the $(d_N + 1)$ -minors of the contraction matrix of a generic value is a polynomial identity of degree at most $d_N + 1 < q^N$ in each variable that holds at all $\mathbb{F}_{q^N}$ -points, so it persists over the algebraic closure $\overline{\mathbb{F}}_q$, and the statement above could be applied over this infinite field.

What the lift must satisfy. Three properties of Ψ are needed on the way.

- (1) The full identity (Pol), not only (Pol1): for $k > q$ the identity (Pw) at $\mathbb{F}_q$ -points does not survive base change (Proposition 14(3)), whereas (Pol) gives (Pw) at every point after base change.
- (2) Alternation rather than antisymmetry, since alternation is what survives base change in characteristic 2.
- (3) An infinite-dimensional image modulo finite rank. For the multipliers $D_a x = ax$ on $L^{\mathbb{N}}$ this holds because the multipliers by the indicators of pairwise disjoint infinite sets are linearly independent modulo finite rank.

The lift of §4 satisfies (Pol) and is alternating (Lemma 25), and its multiplier family is the one in (3), with $L = \kappa$.

4. THE BANACH EXTERIOR TARGET OVER A LAURENT SERIES FIELD

4.1. Setting. Throughout this section κ is a field, $r \in (0, 1)$, and $K_1 := \kappa((X))$ is the field of Laurent series with the absolute value $|a| := r^{\text{ord } a}$ ($|0| = 0$). It is a complete, nontrivially normed, ultrametric field, and $|c| = 1$ for $c \in \kappa^\times$. For $a \in K_1$ and $n \in \mathbb{Z}$ let $\text{coeff}_n(a) \in \kappa$ be its X^n -coefficient. For a set S and $y \in \ell^\infty(S, K_1)$ we also write

$$\text{coeff}_0(y) := (\text{coeff}_0(y(s)))_{s \in S} \in \kappa^S$$

for the constant coefficient taken coordinatewise. This *constant-coefficient map* coeff_0 is κ -linear but not K_1 -linear. Since $|a| < 1$ means $\text{ord } a \geq 1$,

$$(4.1) \quad \|y\|_\infty < 1 \implies \text{coeff}_0(y) = 0, \quad \text{hence} \quad \|y - y'\|_\infty < 1 \implies \text{coeff}_0(y) = \text{coeff}_0(y').$$

Put $E_1 := \ell^\infty(\mathbb{N}, K_1)$ with the sup norm, an ultrametric K_1 -Banach space, and let $E_0 := \kappa^{\mathbb{N}} \subseteq E_1$ be the κ -subspace of sequences with entries in κ . Every nonzero $z \in E_0$ has $\|z\|_\infty = 1$. For $x \in E_1$ and $n \in \mathbb{Z}$ let $x_{[n]} := (\text{coeff}_n(x(i)))_{i \in \mathbb{N}} \in E_0$ be the n -th Laurent component. For $x \neq 0$ the set $\{\text{ord } x(i) : x(i) \neq 0\}$ is bounded below, because $r^{\text{ord } x(i)} \leq \|x\|_\infty$, so it has a minimum $\nu(x)$; then $\|x\|_\infty = r^{\nu(x)}$ and $x_{[n]} = 0$ for $n < \nu(x)$. Finally put

$$M_r := \max_{l \geq 0} (l+1)r^l.$$

The maximum exists because $(l+1)r^l \rightarrow 0$. For $r \leq \frac{1}{2}$ one has $(l+1)r^l \leq (l+1)2^{-l} \leq 1$, so $M_r = 1$.

4.2. The projective exterior norm. On $\Lambda := \Lambda_{K_1}^k E_1$ put

$$\|\omega\|_\pi := \inf \left\{ \sum_j \prod_{i=1}^k \|x_{j,i}\|_\infty : \omega = \sum_j x_{j,1} \wedge \cdots \wedge x_{j,k} \right\},$$

the infimum over finite decompositions, with an ordinary sum. Let $\Omega^{K_1} : \Lambda \rightarrow \ell^\infty(\mathbb{N}^k, K_1)$ be the determinant array of §2.3 with $L = K_1$, $S = \mathbb{N}$ and $V = E_1$, so that $\Omega^{K_1}(x_1 \wedge \cdots \wedge x_k)(i_1, \dots, i_k) = \det(x_b(i_a))_{a,b}$; its values are bounded by the next lemma.

Lemma 23. $\|\cdot\|_\pi$ is a K_1 -norm on Λ with $\|x_1 \wedge \cdots \wedge x_k\|_\pi \leq \prod_i \|x_i\|_\infty$, and Ω^{K_1} is K_1 -linear and injective with $\|\Omega^{K_1}(\omega)\|_\infty \leq \|\omega\|_\pi$. Let B be the completion of $(\Lambda, \|\cdot\|_\pi)$, a K_1 -Banach space, $J : B \rightarrow \ell^\infty(\mathbb{N}^k, K_1)$ the continuous extension of Ω^{K_1} (so $\|J\| \leq 1$), and $W_B : E_1^k \rightarrow B$, $W_B(x) := x_1 \wedge \cdots \wedge x_k$. Then $W_B \in \text{Alt}_{K_1}^k(E_1; B)$, $\|W_B\| \leq 1$, and $J(W_B(x)) = \Omega^{K_1}(x_1 \wedge \cdots \wedge x_k)$.

Proof. Each entry of $\Omega^{K_1}(x_1 \wedge \cdots \wedge x_k)$ is a signed sum of products $x_1(i_{\sigma(1)}) \cdots x_k(i_{\sigma(k)})$, each of absolute value at most $\prod_b \|x_b\|_\infty$; by the ultrametric inequality in K_1 the entry has absolute value at most $\prod_b \|x_b\|_\infty$. For a sum of decomposables, $\|\Omega^{K_1}(\omega)\|_\infty \leq \max_j \prod_i \|x_{j,i}\|_\infty \leq \sum_j \prod_i \|x_{j,i}\|_\infty$, and taking the infimum gives $\|\Omega^{K_1}(\omega)\|_\infty \leq \|\omega\|_\pi$. Injectivity is Lemma 9 with $L = K_1$, $S = \mathbb{N}$ and $V = E_1$. For $\lambda \neq 0$, rescaling the first factor of every wedge gives $\|\lambda\omega\|_\pi \leq |\lambda| \|\omega\|_\pi$, and applying this to λ^{-1} gives equality; concatenating decompositions gives the triangle inequality. Positivity: $\|\omega\|_\pi \geq \|\Omega^{K_1}(\omega)\|_\infty > 0$ for $\omega \neq 0$. Finally W_B is K_1 -multilinear, vanishes when two arguments coincide, and $\|W_B(x)\| \leq \prod_i \|x_i\|_\infty$. $\square$

B is not ultrametric (Proposition 41).

4.3. The support estimate. Let $\Omega^\kappa: \Lambda_\kappa^k E_0 \rightarrow \kappa^{\mathbb{N}^k}$ be the determinant array over κ ; it is injective by Lemma 9 with $L = \kappa$ and $V = E_0$. For $y_1, \dots, y_k \in E_0$ we have $\Omega^{K_1}(y_1 \wedge \cdots \wedge y_k) = \Omega^\kappa(y_1 \wedge \cdots \wedge y_k)$: the determinants are the same, with entries in $\kappa \subseteq K_1$. Support dimensions of elements of $\Lambda_\kappa^k E_0$ are computed in E_0 over κ .

Proposition 24 (support estimate). *For every $b \in B$ there is a unique $\eta(b) \in \Lambda_\kappa^k E_0$ with $\Omega^\kappa(\eta(b)) = \text{coeff}_0(Jb)$. The map η is κ -linear, and*

$$(4.2) \quad \text{sdim}(\eta(b)) \leq kM_r \|b\|_B \quad \text{for all } b \in B.$$

Proof. Step 1: one decomposable. Let $x_1, \dots, x_k \in E_1$. If some $x_b = 0$ the wedge is 0 and $\eta = 0$ works. Otherwise put $\nu_b := \nu(x_b)$ and $l := -\sum_b \nu_b$, so that $\prod_b \|x_b\|_\infty = r^{-l}$. Fix $(i_1, \dots, i_k) \in \mathbb{N}^k$ and write $x_b(i_a) = \sum_{n \geq \nu_b} \text{coeff}_n(x_b(i_a)) X^n$. Expand $\det(x_b(i_a))_{a,b} = \sum_\sigma \text{sgn}(\sigma) \prod_b x_b(i_{\sigma(b)})$. The factor $x_b(i_{\sigma(b)})$ has order at least ν_b , so the constant coefficient of the product is the finite sum $\sum_n \prod_b \text{coeff}_{n_b}(x_b(i_{\sigma(b)}))$ over

$$\mathcal{N} := \left\{ n \in \mathbb{Z}^k : n_b \geq \nu_b \text{ for all } b, \sum_b n_b = 0 \right\}.$$

Exchanging the two finite sums,

$$\text{coeff}_0 \det(x_b(i_a))_{a,b} = \sum_{n \in \mathcal{N}} \det(x_{b,[n_b]}(i_a))_{a,b},$$

where $x_{b,[n]}$ is the n -th Laurent component of x_b . Hence $\text{coeff}_0(\Omega^{K_1}(x_1 \wedge \cdots \wedge x_k)) = \Omega^\kappa(\eta)$ with

$$\eta := \sum_{n \in \mathcal{N}} x_{1,[n_1]} \wedge \cdots \wedge x_{k,[n_k]} \in \Lambda_\kappa^k E_0.$$

If $l < 0$, then $\sum_b n_b \geq \sum_b \nu_b = -l > 0$ for all n with $n_b \geq \nu_b$, so $\mathcal{N} = \emptyset$ and $\eta = 0$: a decomposable of small norm has zero constant coefficient. If $l \geq 0$, every $n \in \mathcal{N}$ has $n_b - \nu_b \geq 0$ and $\sum_b (n_b - \nu_b) = l$, so $\nu_b \leq n_b \leq \nu_b + l$. Then η is a sum of wedges of vectors from the set $\{x_{b,[n]} : 1 \leq b \leq k, \nu_b \leq n \leq \nu_b + l\}$ of at most $k(l+1)$ vectors, and

$$\text{sdim}(\eta) \leq k(l+1) = k(l+1)r^l \cdot r^{-l} \leq kM_r \prod_b \|x_b\|_\infty.$$

In all cases $\text{sdim}(\eta) \leq kM_r \prod_b \|x_b\|_\infty$.

Step 2: an element of Λ . Let $\omega = \sum_j w_j$ with $w_j = x_{j,1} \wedge \cdots \wedge x_{j,k}$, and let η_j be as in Step 1 for w_j . Since coeff_0 and Ω^{K_1} are additive, $\text{coeff}_0(\Omega^{K_1}(\omega)) = \sum_j \text{coeff}_0(\Omega^{K_1}(w_j)) = \Omega^\kappa(\sum_j \eta_j)$. By injectivity of Ω^κ the element $\eta(\omega) := \sum_j \eta_j$ does not depend on the decomposition, and $\text{sdim}(\eta(\omega)) \leq \sum_j \text{sdim}(\eta_j) \leq kM_r \sum_j \prod_i \|x_{j,i}\|_\infty$. Taking the infimum over decompositions gives $\text{sdim}(\eta(\omega)) \leq kM_r \|\omega\|_\pi$. (The infimum need not be attained.)

Step 3: completion. Let $b \in B$ and $\omega_n \in \Lambda$ with $\|b - \omega_n\|_B \rightarrow 0$. Then $\|Jb - \Omega^{K_1}(\omega_n)\|_\infty \leq \|b - \omega_n\|_B < 1$ for $n \geq n_0$, so by (4.1) $\text{coeff}_0(Jb) = \text{coeff}_0(\Omega^{K_1}(\omega_n)) = \Omega^\kappa(\eta(\omega_n))$ for $n \geq n_0$. Put $\eta(b) := \eta(\omega_n)$, which is independent of $n \geq n_0$ by injectivity of Ω^κ . Then $\text{sdim}(\eta(b)) \leq kM_r\|\omega_n\|_\pi$ for all $n \geq n_0$, and $\|\omega_n\|_\pi \rightarrow \|b\|_B$ gives (4.2). Uniqueness is injectivity of Ω^κ , and η is κ -linear because J is K_1 -linear, coeff_0 is κ -linear and Ω^κ is injective and linear. $\square$

4.4. The coefficient lift. Suppose that P is a bounded k -linear lift of $Q_{E_1, E_1, B}^{K_1}$ over K_1 . For $a \in E_1$ let $D_a \in \mathcal{L}(E_1, E_1)$ be coordinatewise multiplication, $D_a x = ax$. The map $a \mapsto D_a$ is K_1 -linear and $\|D_a\| \leq \|a\|_\infty$. Define

$$\Phi(a_1, \dots, a_k; x_1, \dots, x_k) := P(D_{a_1}, \dots, D_{a_k})(W_B)(x_1, \dots, x_k) \in B,$$

$$\Psi := \eta \circ \Phi|_{E_0^{2k}} : E_0^k \times E_0^k \rightarrow \Lambda_\kappa^k E_0.$$

Since $\Omega^\kappa \circ \eta = \text{coeff}_0 \circ J$, the map Ψ is $\text{coeff}_0 \circ J \circ \Phi|_{E_0^{2k}}$, read in $\Lambda_\kappa^k E_0$ through the injective map Ω^κ .

Lemma 25. *The map Ψ has the following properties.*

- ($\Psi 1$) Ψ is $2k$ -linear over κ .
- ($\Psi 2$) Ψ is alternating in the last k slots.
- ($\Psi 3$) $\text{sdim}(\Psi(z)) \leq d := \lfloor kM_r\|P\| \rfloor$ for all $z \in E_0^{2k}$.
- ($\Psi 4$) Ψ satisfies (Pol) for the multiplier family $D^\kappa : E_0 \rightarrow \text{End}_\kappa(E_0)$, $D_a^\kappa x = ax$: for all $m \geq 1$, $b_1, \dots, b_m \in E_0$, $\alpha \in \mathbb{N}^m$ with $|\alpha| = k$, and $x \in E_0^k$,

$$\sum_{\text{type } f = \alpha} \Psi(b_f; x) = \sum_{\text{type } f = \alpha} (b_{f(1)}x_1) \wedge \dots \wedge (b_{f(k)}x_k) \quad \text{in } \Lambda_\kappa^k E_0.$$

In particular Ψ satisfies (Pol1).

Proof. ($\Psi 1$) Φ is K_1 -multilinear, hence κ -multilinear on the κ -subspace E_0 , and η is κ -linear. ($\Psi 2$) $P(\dots)(W_B)$ lies in $\text{Alt}_{K_1}^k(E_1; B)$, and η is linear. ($\Psi 3$) Inputs from E_0 have norm at most 1, so $\|\Phi(a; x)\| \leq \|P\| \prod_i \|D_{a_i}\| \|W_B\| \prod_i \|x_i\|_\infty \leq \|P\|$; now use (4.2).

($\Psi 4$) Fix $b_1, \dots, b_m \in E_0$ and $x \in E_0^k$, and for $\lambda \in K_1^m$ put $a_\lambda := \sum_i \lambda_i b_i \in E_1$. The diagonal over K_1 : the lift property gives, for every $\lambda \in K_1^m$,

$$\Phi(a_\lambda, \dots, a_\lambda; x) = P(D_{a_\lambda}, \dots, D_{a_\lambda})(W_B)(x) = Q(D_{a_\lambda})(W_B)(x) = W_B(a_\lambda x_1, \dots, a_\lambda x_k).$$

Expansion: expanding both sides by multilinearity gives $\sum_\alpha \lambda^\alpha (L_\alpha - R_\alpha) = 0$ for all $\lambda \in K_1^m$, where

$$L_\alpha := \sum_{\text{type } f = \alpha} \Phi(b_f; x) \in B, \quad R_\alpha := \sum_{\text{type } f = \alpha} W_B(b_{f(1)}x_1, \dots, b_{f(k)}x_k) \in B.$$

Identity principle: K_1 is infinite even when κ is finite, so Lemma 12 in the K_1 -vector space B gives $L_\alpha = R_\alpha$ for every α . Apply η : η is additive, so $\eta(L_\alpha) = \sum_{\text{type } f = \alpha} \Psi(b_f; x)$. For $y_1, \dots, y_k \in E_0$, Lemma 23 and the remark before Proposition 24 give $J(W_B(y)) = \Omega^{K_1}(y_1 \wedge \dots \wedge y_k) = \Omega^\kappa(y_1 \wedge \dots \wedge y_k)$; this array has entries in κ , so coeff_0 fixes it, and uniqueness in Proposition 24 gives $\eta(W_B(y)) = y_1 \wedge \dots \wedge y_k \in \Lambda_\kappa^k E_0$. With $y_j = b_{f(j)}x_j \in E_0$ this yields $\eta(R_\alpha) = \sum_{\text{type } f = \alpha} (b_{f(1)}x_1) \wedge \dots \wedge (b_{f(k)}x_k)$. $\square$

The polarization in ($\Psi 4$) takes place over the infinite field K_1 , before the κ -linear map η is applied. Evaluating the diagonal only at κ -points would give (Pw) at κ -points, which for $\kappa = \mathbb{F}_q$ and $k > q$ is strictly weaker than (Pol) (Proposition 14(3)); for the proof below (Pol1) suffices.

4.5. No bounded lift over a Laurent series field.

Theorem 26. *Let κ be a finite field of characteristic p , let $k \geq p$ and $r \in (0, 1)$, and let $K_1 = \kappa((X))$ with $|X| = r$. Let $E_1 = \ell^\infty(\mathbb{N}, K_1)$, and let B be the completion of $\Lambda_{K_1}^k E_1$ under the projective exterior norm. Then $Q_{E_1, E_1, B}^{K_1}$ admits no bounded k -linear lift over K_1 . Consequently it is analytic at no point of $\mathcal{L}(E_1, E_1)$.*

Proof. Suppose P exists, and form Ψ as in §4.4. By Lemma 25, $\Psi: E_0^k \times E_0^k \rightarrow \Lambda_\kappa^k E_0$ with $E_0 = \kappa^\mathbb{N}$ is $2k$ -linear over κ , alternating and hence antisymmetric in the last k slots, satisfies (Pol1), and has $\text{sdim}(\Psi(z)) \leq d$ for all inputs z . Moreover $k! = 0$ in κ , since $k \geq p = \text{char } \kappa$. This contradicts Theorem 15(2) with $L = \kappa$ and $E_0 = \kappa^\mathbb{N}$. The last sentence follows from Proposition 6. $\square$

Remark 27. Lemmas 23 and 25 and Proposition 24 hold for every field κ , and nothing in them uses a normalization of r : the constant M_r makes the argument uniform in $r \in (0, 1)$. Only the final appeal to Theorem 15 needs κ finite. We do not need, and do not discuss, the analogue of Theorem 26 for infinite κ with this particular pair (E_1, B) . For an infinite κ of characteristic p the field $\kappa((X))$ is covered by Theorem 1, with the pair obtained by ascent from its subfield $\mathbb{F}_p((X))$.

5. ASCENT TO AN ARBITRARY BASE FIELD

5.1. The Laurent subfield. A metric space is *spherically complete* if every nonempty family of closed balls, any two of which meet, has a common point.

Lemma 28 (discrete distances). *Let Y be a complete ultrametric space whose distances lie in $r^\mathbb{Z} \cup \{0\}$ for some $r \in (0, 1)$. Then Y is spherically complete.*

Proof. We use two facts about closed balls $\mathcal{D}(a, \rho) = \{y \in Y : d(y, a) \leq \rho\}$ in an ultrametric space: every point of a ball is a center, and hence, if two closed balls meet, the one with the smaller radius is contained in the other. Since all distances lie in $r^\mathbb{Z} \cup \{0\}$, we have $\mathcal{D}(a, \rho) = \mathcal{D}(a, \rho^b)$ with $\rho^b := \max\{s \in r^\mathbb{Z} \cup \{0\} : s \leq \rho\}$, so we may assume that all radii lie in $r^\mathbb{Z} \cup \{0\}$. Let $(\mathcal{D}(a_i, \rho_i))_{i \in I}$ be a nonempty family of pairwise meeting closed balls. A subset of $r^\mathbb{Z} \cup \{0\}$ has a least element unless its infimum is 0 and not attained. If some ρ_{i_0} is least, then $\mathcal{D}(a_{i_0}, \rho_{i_0}) \subseteq \mathcal{D}(a_i, \rho_i)$ for every i , and a_{i_0} is a common point. Otherwise all $\rho_i > 0$ and there are $i_1, i_2, \dots$ with ρ_{i_n} strictly decreasing to 0. The balls $\mathcal{D}(a_{i_n}, \rho_{i_n})$ are nested, so $d(a_{i_{n'}}, a_{i_n}) \leq \rho_{i_n}$ for $n' \geq n$; the centers form a Cauchy sequence with a limit a , which lies in every $\mathcal{D}(a_{i_n}, \rho_{i_n})$. For any i choose n with $\rho_{i_n} \leq \rho_i$; then $\mathcal{D}(a_{i_n}, \rho_{i_n}) \subseteq \mathcal{D}(a_i, \rho_i)$, so a is a common point. $\square$

Remark 29 (the hypothesis (K2) of [6]). The construction of [6] assumes that the base field satisfies the hypothesis (K2) there: there are closed balls $\mathcal{D}(c_n, r_n)$ with $|c_{n+1} - c_n| \leq r_n$, $r_0 > r_1 > \dots$ and $\bigcap_n \mathcal{D}(c_n, r_n) = \emptyset$. An ultrametric field (indeed any ultrametric space) satisfies (K2) if and only if it is not spherically complete. Indeed, the balls in (K2) pairwise meet, since $|c_n - c_m| \leq \max_{m \leq j < n} |c_{j+1} - c_j| \leq r_m$ for $m \leq n$. Conversely, let a family of pairwise meeting closed balls have no common point. A ball of least radius in the family would lie in all the others (as in the proof of Lemma 28), and its center would be a common point; so the radii have no least element, and every radius in the family exceeds their infimum. A sequence of balls from the family whose radii decrease strictly to that infimum is then nested, so it satisfies $|c_{n+1} - c_n| \leq r_n$, and it has empty intersection: a common point of the sequence would lie in every ball of the family, because every ball of the family contains a ball of the sequence. This remark is used only in §1 and §8.

Lemma 30 (Laurent subfield). *Let K' be a complete nontrivially normed field of characteristic p , and let $t \in K'$ with $0 < |t| = r < 1$.*

- (1) K' is ultrametric, and $|c| = 1$ for every $c \in \mathbb{F}_p^\times$.
- (2) For $a_m, \dots, a_M \in \mathbb{F}_p$, not all zero, $|\sum_{i=m}^M a_i t^i| = r^{\min\{i: a_i \neq 0\}}$.
- (3) Give $\mathbb{F}_p((X))$ the absolute value with $|X| = r$. There is a continuous isometric ring homomorphism $\text{ev}_t: \mathbb{F}_p((X)) \rightarrow K'$ such that $\text{ev}_t(\sum_{i \geq m} a_i X^i) = \sum_{i \geq m} a_i t^i$. Its image K_1 is a closed subfield of K' ; it is the closure of $\mathbb{F}_p(t)$, and it is isometrically isomorphic to $\mathbb{F}_p((X))$.
- (4) K_1 is spherically complete.

Proof. (1) K' is ultrametric by Lemma 4, and $c^{p-1} = 1$ gives $|c| = 1$.

(2) By (1) the nonzero terms have pairwise distinct absolute values r^i . In an ultrametric field $|x + y| = |x|$ whenever $|y| < |x|$, so the absolute value of the sum is the largest one, $r^{\min\{i: a_i \neq 0\}}$.

(3) On Laurent polynomials, $\text{ev}_t: \mathbb{F}_p[X, X^{-1}] \rightarrow K'$ is a ring homomorphism with $|\text{ev}_t(g)| = r^{\text{ord } g} = |g|$, by (2) after multiplying by a power of t . Laurent polynomials are dense in $\mathbb{F}_p((X))$ (truncations converge), and K' is complete, so ev_t extends uniquely to a continuous map on $\mathbb{F}_p((X))$. The extension is isometric, additive and multiplicative by continuity, and it maps the partial sums of $\sum a_i X^i$ to those of $\sum a_i t^i$. The image of the complete space $\mathbb{F}_p((X))$ under an isometry is complete, hence closed; it is a subring, and a subfield because $\text{ev}_t(g) \text{ev}_t(g^{-1}) = 1$. It contains $\mathbb{F}_p(t)$, and $\mathbb{F}_p[t, t^{-1}] \subseteq \mathbb{F}_p(t)$ is dense in it; so it is the closure of $\mathbb{F}_p(t)$.

(4) K_1 is complete by (3) and ultrametric by (1), and $|K_1^\times| = r^\mathbb{Z}$, so its distances lie in $r^\mathbb{Z} \cup \{0\}$. Apply Lemma 28. $\square$

5.2. Ingleton's theorem and its consequences. Throughout the rest of this section we fix fields $K_1 \subseteq K'$ with the following properties.

- (H1) K_1 , with the restricted absolute value, is nontrivially normed, complete and spherically complete.
- (H2) K' is ultrametric.

By multiplication, K' is an ultrametric normed K_1 -space. In our application $K' = \widehat{K}$ and K_1 is the field of Lemma 30.

Lemma 31. *Let $\mathbb{K}$ be a complete nontrivially normed field and U a finite-dimensional normed $\mathbb{K}$ -space. Then every linear functional on U is bounded, and U is complete.*

Proof. By induction on $n = \dim U$; the case $n = 0$ is trivial. Let $n \geq 1$ and let $g \neq 0$ be a linear functional on U . Its kernel H has dimension $n - 1$, so it is complete by induction, hence closed in U . Choose v with $g(v) = 1$; then $\delta := \inf_{h \in H} \|v - h\| > 0$. Every $x \in U$ is $x = g(x)v + h$ with $h \in H$, and if $g(x) \neq 0$ then $\|x\| = |g(x)| \|v + g(x)^{-1}h\| \geq |g(x)|\delta$. So $|g(x)| \leq \delta^{-1}\|x\|$. For completeness, the coordinate map of a basis $U \rightarrow \mathbb{K}^n$ (max norm) is bounded by the first part, and its inverse $(c_i) \mapsto \sum_i c_i e_i$ is bounded by $\sum_i \|e_i\|$; so U is bi-Lipschitz equivalent to the complete space $\mathbb{K}^n$. $\square$

Theorem 32 (Ingleton [3]; see also [10]). *Let K_1 be a spherically complete ultrametric normed field, V a K_1 -vector space with an ultrametric norm, $V_0 \subseteq V$ a subspace, $C \geq 0$, and $T_0: V_0 \rightarrow K_1$ linear with $|T_0 v| \leq C\|v\|$ for $v \in V_0$. Then T_0 extends to a linear map $T: V \rightarrow K_1$ with $|Tx| \leq C\|x\|$ for all $x \in V$.*

We apply Theorem 32 only with the ultrametric domain $V = K'$ and the codomain K_1 . It is never applied to the spaces E_1, F_1 below, which need not be ultrametric.

Corollary 33 (Ingleton projection). *There is a K_1 -linear map $\varpi: K' \rightarrow K_1$ with $\varpi|_{K_1} = \text{id}$ and $|\varpi(\lambda)| \leq |\lambda|$ for all $\lambda \in K'$.*

Proof. Theorem 32 with $V = K'$, $V_0 = K_1$, $T_0 = \text{id}$ and $C = 1$; its hypotheses hold by (H1) and (H2). $\square$

Lemma 34 (extension). *Let $\mathbb{K} \in \{K_1, K'\}$.*

- (1) *Let V_0 be a dense $\mathbb{K}$ -subspace of a normed $\mathbb{K}$ -space V , let Y be a $\mathbb{K}$ -Banach space and $T_0: V_0 \rightarrow Y$ a $\mathbb{K}$ -linear map with $\|T_0x\| \leq C\|x\|$. Then T_0 extends uniquely to a continuous map $T: V \rightarrow Y$, which is $\mathbb{K}$ -linear with $\|Tx\| \leq C\|x\|$.*
- (2) *Let $V_1, \dots, V_k$ be normed K' -spaces with dense subspaces V_i^0 , Y a K' -Banach space, and $M: \prod_i V_i^0 \rightarrow Y$ a K' -multilinear map with $\|M(d)\| \leq C \prod_i \|d_i\|$. Then M extends uniquely to a continuous K' -multilinear map $\bar{M}: \prod_i V_i \rightarrow Y$ with $\|\bar{M}\| \leq C$.*

Proof. (1) T_0 is uniformly continuous and Y is complete, so it extends uniquely to a continuous map; linearity and the bound pass to limits by continuity of addition and of scalar multiplication. (2) For $d, d' \in \prod_i V_i^0$ with all $\|d_i\|, \|d'_i\| \leq \rho$ the telescoping estimate $\|M(d) - M(d')\| \leq C\rho^{k-1} \sum_i \|d_i - d'_i\|$ holds. So M is uniformly continuous on bounded subsets of $\prod_i V_i^0$, and it extends uniquely to a continuous map on their closures, which cover $\prod_i V_i$; multilinearity and the bound pass to the limit. $\square$

5.3. Projective base change. For a normed K_1 -space V let $V_{K'} := V \otimes_{K_1} K'$, a K' -vector space through the second factor, and put

$$\|u\|_\pi := \inf \left\{ \sum_j \|x_j\| |\lambda_j| : u = \sum_j x_j \otimes \lambda_j \right\},$$

the infimum over finite decompositions, with an ordinary sum. For a bounded K_1 -linear $\psi: K' \rightarrow K_1$ let $T_\psi: V_{K'} \rightarrow V$ be the K_1 -linear map with $T_\psi(x \otimes \lambda) = \psi(\lambda)x$; it exists because $(x, \lambda) \mapsto \psi(\lambda)x$ is K_1 -bilinear. Let $\iota_V: V \rightarrow V_{K'}$, $x \mapsto x \otimes 1$; it is K_1 -linear, since $cx \otimes 1 = x \otimes c = c \cdot (x \otimes 1)$ for $c \in K_1$.

Lemma 35 (projective base change). *Let V be a normed K_1 -space; no ultrametric hypothesis is made on V .*

- (1) $\|T_\psi u\| \leq \|\psi\| \|u\|_\pi$ for every bounded K_1 -linear $\psi: K' \rightarrow K_1$.
- (2) $\|\cdot\|_\pi$ is a K' -norm on $V_{K'}$.
- (3) ι_V is isometric.
- (4) Let $V \hat{\otimes}_\pi K'$ be the completion of $(V_{K'}, \|\cdot\|_\pi)$. If V is complete, then T_ϖ , with ϖ as in Corollary 33, extends to a K_1 -linear map $\Pi_V: V \hat{\otimes}_\pi K' \rightarrow V$ with $\|\Pi_V\| \leq 1$ and $\Pi_V \circ \iota_V = \text{id}_V$.

Proof. (1) For any decomposition $u = \sum_j x_j \otimes \lambda_j$ we have

$$\|T_\psi u\| = \left\| \sum_j \psi(\lambda_j) x_j \right\| \leq \sum_j \|\psi\| |\lambda_j| \|x_j\|;$$

take the infimum.

(2) Multiplying the second factors by $\mu \in K'$ gives $\|\mu u\|_\pi \leq |\mu| \|u\|_\pi$, and applying this to μ^{-1} gives equality for $\mu \neq 0$; concatenating decompositions gives the triangle inequality. For positivity let $u \neq 0$ and write $u = \sum_{j=1}^n x_j \otimes \lambda_j$ with $\lambda_1, \dots, \lambda_n$ linearly independent over K_1 (rewrite any representation in a basis of the K_1 -span of its second factors). Since $u \neq 0$, some $x_{j_0} \neq 0$. The K_1 -span U of the λ_j is a finite-dimensional normed K_1 -space, so the coordinate functional $\psi_0: U \rightarrow K_1$, $\psi_0(\lambda_j) = \delta_{jj_0}$, is bounded by some C (Lemma 31; K_1 is complete by

(H1)). By Theorem 32, applied to the ultrametric K_1 -space K' , it extends to $\psi: K' \rightarrow K_1$ with $\|\psi\| \leq C$. Then $T_\psi u = x_{j_0}$, and (1) gives $0 < \|x_{j_0}\| \leq C\|u\|_\pi$. No functional on V is used.

(3) $\|x \otimes 1\|_\pi \leq \|x\|$, and $\|x\| = \|T_\varpi(x \otimes 1)\| \leq \|x \otimes 1\|_\pi$ by (1), since $\|\varpi\| \leq 1$ and $\varpi(1) = 1$.

(4) By (1), T_ϖ is K_1 -linear and 1-Lipschitz on the dense subspace $V_{K'}$, with values in the complete space V ; apply Lemma 34(1) with $\mathbb{K} = K_1$. Finally $\Pi_V \iota_V x = T_\varpi(x \otimes 1) = \varpi(1)x = x$. $\square$

The data. Let E_1 be a normed K_1 -space and F_1 a K_1 -Banach space. Put

$$E := E_1 \widehat{\otimes}_\pi K', \quad F := F_1 \widehat{\otimes}_\pi K',$$

the completions of $((E_1)_{K'}, \|\cdot\|_\pi)$ and $((F_1)_{K'}, \|\cdot\|_\pi)$. They are K' -Banach spaces; $\iota_E: E_1 \rightarrow E$ and $\iota_F: F_1 \rightarrow F$ are K_1 -linear isometries, and $\Pi_F: F \rightarrow F_1$ is K_1 -linear with $\|\Pi_F\| \leq 1$ and $\Pi_F \circ \iota_F = \text{id}_{F_1}$ (Lemma 35).

Lemma 36 (operators). *For $f \in \mathcal{L}_{K_1}(E_1, E_1)$ the map $f \otimes \text{id}$ on $(E_1)_{K'}$ is K' -linear with $\|(f \otimes \text{id})u\|_\pi \leq \|f\| \|u\|_\pi$. Its extension $f_{K'} \in \mathcal{L}_{K'}(E, E)$ satisfies $\|f_{K'}\| \leq \|f\|$ and $f_{K'} \circ \iota_E = \iota_E \circ f$, and $f \mapsto f_{K'}$ is K_1 -linear.*

Proof. Apply f to the first factors of any decomposition, and extend by Lemma 34(1) with $\mathbb{K} = K'$, since E is complete. The identity $f_{K'} \iota_E = \iota_E f$ and the K_1 -linearity in f are visible on elementary tensors. $\square$

Lemma 37 (forms). *For $m \in \text{Alt}_{K_1}^k(E_1; F_1)$ there is $m_{K'} \in \text{Alt}_{K'}^k(E; F)$ with $\|m_{K'}\| \leq \|m\|$ and*

$$m_{K'}(\iota_E x_1, \dots, \iota_E x_k) = \iota_F(m(x_1, \dots, x_k)) \quad (x_i \in E_1),$$

and $m \mapsto m_{K'}$ is K_1 -linear.

Proof. Fix a K_1 -basis $(\lambda_\beta)_{\beta \in \mathfrak{B}}$ of K' . Every $u \in (E_1)_{K'}$ has a unique expansion $u = \sum_\beta u^\beta \otimes \lambda_\beta$ with $u^\beta \in E_1$, finitely many nonzero, and $u \mapsto u^\beta$ is K_1 -linear. Define

$$m^{\text{alg}}(u_1, \dots, u_k) := \sum_{\beta_1, \dots, \beta_k} m(u_1^{\beta_1}, \dots, u_k^{\beta_k}) \otimes \lambda_{\beta_1} \cdots \lambda_{\beta_k} \in (F_1)_{K'}.$$

(i) m^{alg} is K_1 -multilinear.

(ii) $m^{\text{alg}}(x_1 \otimes \mu_1, \dots, x_k \otimes \mu_k) = m(x_1, \dots, x_k) \otimes \mu_1 \cdots \mu_k$ for $x_i \in E_1$ and $\mu_i \in K'$. Indeed, write $\mu_i = \sum_\beta c_{i\beta} \lambda_\beta$ with $c_{i\beta} \in K_1$; then $x_i \otimes \mu_i = \sum_\beta (c_{i\beta} x_i) \otimes \lambda_\beta$ and $m^{\text{alg}} = \sum_\beta \prod_i c_{i\beta_i} m(x) \otimes \prod_i \lambda_{\beta_i} = m(x) \otimes \prod_i (\sum_\beta c_{i\beta} \lambda_\beta)$.

(iii) m^{alg} is K' -multilinear. By (i) and additivity it suffices to check K' -homogeneity in one slot when every slot is an elementary tensor, and there (ii) gives it.

(iv) $\|m^{\text{alg}}(u_1, \dots, u_k)\|_\pi \leq \|m\| \prod_i \|u_i\|_\pi$. For arbitrary decompositions $u_i = \sum_j x_{ij} \otimes \mu_{ij}$, multilinearity and (ii) give $m^{\text{alg}}(u) = \sum_{j_1, \dots, j_k} m(x_{1j_1}, \dots, x_{kj_k}) \otimes \prod_i \mu_{ij_i}$, of projective norm at most $\|m\| \prod_i (\sum_j \|x_{ij}\| \|\mu_{ij}\|)$; take the infimum in each slot.

(v) m^{alg} is alternating in the strong sense. Let slots $a \neq b$ both carry $u = \sum_j x_j \otimes \mu_j$, and expand every slot into elementary tensors. By (ii) each term is $m(\dots) \otimes (\text{scalar})$. Terms with the same index j in slots a and b contain m with a repeated argument and vanish. The terms with indices (j, j') , $j \neq j'$, in slots (a, b) pair off with those with indices (j', j) : the scalars agree, and the values of m are negatives of each other because m is antisymmetric. So the total is 0. This argument is valid in every characteristic, including 2; checking only the full diagonal $m^{\text{alg}}(u, \dots, u) = 0$ would not suffice for $k \geq 3$.

(vi) By (iii), (iv) and Lemma 34(2) (F is complete), m^{alg} extends to a continuous K' -multilinear map $m_{K'}: E^k \rightarrow F$ with $\|m_{K'}\| \leq \|m\|$. It is alternating: if $v \in E^k$ has $v_a = v_b$, approximate $v_a = v_b$ by a single sequence in $(E_1)_{K'}$ and the other v_i by sequences in $(E_1)_{K'}$; m^{alg} vanishes along the approximants by (v), so $m_{K'}(v) = 0$ by continuity.

(vii) $m_{K'}(\iota_E x_1, \dots, \iota_E x_k) = m^{\text{alg}}(x_1 \otimes 1, \dots, x_k \otimes 1) = m(x) \otimes 1 = \iota_F(m(x))$ by (ii). K_1 -linearity of $m \mapsto m^{\text{alg}}$ is visible in the formula, and it passes to the extensions by uniqueness. $\square$

Proposition 38 (descent of a lift). *Assume (H1) and (H2). Let E_1 be a normed K_1 -space, F_1 a K_1 -Banach space, and E, F as above. If $Q_{E,E,F}^{K'}$ admits a bounded k -linear lift P over K' , then $Q_{E_1,E_1,F_1}^{K_1}$ admits a bounded k -linear lift P_1 over K_1 with $\|P_1\| \leq \|P\|$. Equivalently, by Proposition 6 over K_1 and over K' : if $Q_{E_1,E_1,F_1}^{K_1}$ is analytic at no point, then $Q_{E,E,F}^{K'}$ is analytic at no point either.*

Proof. For $f_1, \dots, f_k \in \mathcal{L}_{K_1}(E_1, E_1)$, $m \in \text{Alt}_{K_1}^k(E_1; F_1)$ and $x \in E_1^k$ define

$$P_1(f_1, \dots, f_k)(m)(x_1, \dots, x_k) := \Pi_F \left(P((f_1)_{K'}, \dots, (f_k)_{K'})(m_{K'})(\iota_E x_1, \dots, \iota_E x_k) \right).$$

It is well defined. $P((f_i)_{K'})(m_{K'}) \in \text{Alt}_{K'}^k(E; F)$ is K' -multilinear, hence K_1 -multilinear, and ι_E and Π_F are K_1 -linear. So $x \mapsto P_1(f)(m)(x)$ is K_1 -multilinear, vanishes when two x_i coincide (then the $\iota_E x_i$ coincide), and has norm at most $\|\Pi_F\| \|P\| \prod_i \|(f_i)_{K'}\| \|m_{K'}\| \leq \|P\| \prod_i \|f_i\| \|m\|$ by Lemmas 35–37. So $P_1(f)(m) \in \text{Alt}_{K_1}^k(E_1; F_1)$. It is K_1 -linear in m , because $m \mapsto m_{K'}$ is K_1 -linear and $P(\dots)$ is K' -linear, and K_1 -multilinear in $(f_1, \dots, f_k)$, because $f \mapsto f_{K'}$ is K_1 -linear and P is K' -multilinear. Hence P_1 is a bounded k -linear map over K_1 with $\|P_1\| \leq \|P\|$.

The diagonal. Using $P(g, \dots, g) = Q^{K'}(g)$ with $g = f_{K'}$, Lemma 36 and Lemma 37,

$$\begin{aligned} P_1(f, \dots, f)(m)(x) &= \Pi_F(m_{K'}(f_{K'} \iota_E x_1, \dots, f_{K'} \iota_E x_k)) = \Pi_F(m_{K'}(\iota_E f x_1, \dots, \iota_E f x_k)) \\ &= \Pi_F(\iota_F(m(f x_1, \dots, f x_k))) = m(f x_1, \dots, f x_k) = Q^{K_1}(f)(m)(x). \quad \square \end{aligned}$$

Remark 39. Proposition 38 is characteristic-free and uses no hypothesis on k . It transports negative results upward only, from K_1 to K' . Ingletton's theorem is used only on K' , so F_1 may be, and in our application is, non-ultrametric. The completeness of F_1 is used essentially: it is what makes the retraction Π_F land in F_1 .

5.4. Incomplete base fields.

Lemma 40. *Let K be a nontrivially normed field with completion $\widehat{K}$, and let E, E', F be $\widehat{K}$ -Banach spaces, regarded as K -Banach spaces by restriction of scalars (with the same norms). Then $\mathcal{L}_K(E, E') = \mathcal{L}_{\widehat{K}}(E, E')$, $\text{Alt}_K^k(E; F) = \text{Alt}_{\widehat{K}}^k(E; F)$ and $\text{Alt}_K^k(E'; F) = \text{Alt}_{\widehat{K}}^k(E'; F)$ with the same norms, $Q_{E,E',F}^K = Q_{E,E',F}^{\widehat{K}}$, and every bounded k -linear lift of $Q_{E,E',F}^K$ over K is a bounded k -linear lift of $Q_{E,E',F}^{\widehat{K}}$ over $\widehat{K}$.*

Proof. A continuous K -linear map g between $\widehat{K}$ -normed spaces is $\widehat{K}$ -linear: for $\lambda \in \widehat{K}$ choose $\lambda_n \in K$ with $\lambda_n \rightarrow \lambda$; then $\lambda_n x \rightarrow \lambda x$ because $\|\lambda_n x - \lambda x\| = |\lambda_n - \lambda| \|x\|$, so $g(\lambda x) = \lim g(\lambda_n x) = \lim \lambda_n g(x) = \lambda g(x)$. Likewise a continuous K -multilinear map is $\widehat{K}$ -multilinear, slot by slot. So the spaces of bounded maps coincide as sets, with the same sup-formula norms, and they are $\widehat{K}$ -Banach spaces. The map Q is given by the same formula over both fields. A bounded k -linear map P over K between these spaces is continuous in each slot, hence k -linear over $\widehat{K}$. $\square$

6. PROOFS OF THE MAIN THEOREM AND THE COROLLARY

We first show that the target has no equivalent ultrametric norm, and then assemble the proofs of Theorem 1 and Corollary 2.

6.1. The target has no equivalent ultrametric norm.

Proposition 41. *Let κ be any field, $k \geq 2$, $r \in (0, 1)$, and let $K_1 = \kappa((X))$, E_1 , E_0 and B be as in §4. Let $e_i \in E_0$ be the unit vectors and $\omega_j := e_{kj+1} \wedge \cdots \wedge e_{k(j+k)} \in B$ ($j \geq 0$), wedges on pairwise disjoint blocks of indices. Then $\|\omega_j\|_B = 1$ and*

$$\frac{N}{M_r} \leq \left\| \sum_{j < N} \omega_j \right\|_B \leq N \quad (N \geq 1).$$

Hence the norm of B is not ultrametric, and B admits no equivalent ultrametric norm. The same holds for $F = B \widehat{\otimes}_\pi K'$ as in §5, for any $K' \supseteq K_1$ satisfying (H1) and (H2), regarded as a normed space over K' or over any subfield of K' .

Proof. The upper bounds follow from the defining decompositions, since unit vectors have norm 1: each ω_j is one wedge of unit vectors, and $\sum_{j < N} \omega_j$ is a sum of N of them. Also $\|\omega_j\|_B \geq \|J\omega_j\|_\infty \geq 1$, because the entry of $J\omega_j$ at the block indices is $\det I = 1$.

Put $\omega := \sum_{j < N} \omega_j$. It is the image of the element $\sum_{j < N} \omega_j$ of $\Lambda_\kappa^k E_0$, again denoted ω ; its array $J\omega = \Omega^\kappa(\omega)$ has entries in κ , so coeff_0 fixes it and $\eta(\omega) = \omega$ by uniqueness in Proposition 24. We bound $\text{sdim}(\omega)$ from below with Lemma 11. Fix a block $j < N$ and an index i_0 in it, and let φ consist of the $k-1$ coordinate functionals $e_{i'}^*$ of the other indices i' of block j . For a block $j' \neq j$ all φ_a vanish on the vectors of $\omega_{j'}$, so every $(k-1) \times (k-1)$ matrix in the formula for $c_\varphi(\omega_{j'})$ is zero and $c_\varphi(\omega_{j'}) = 0$ (here $k \geq 2$ is used). In $c_\varphi(\omega_j)$, the term of the vector e_{i_0} has a permutation matrix, with determinant ± 1 , and every other term has a matrix containing the column of e_{i_0} , on which all φ_a vanish. So $c_\varphi(\omega) = \pm e_{i_0}$. The span of the $c_\varphi(\omega)$ therefore contains the kN unit vectors of the first N blocks, and $\text{sdim}(\omega) \geq kN$. Now (4.2) gives $kN \leq \text{sdim}(\eta(\omega)) \leq kM_r \|\omega\|_B$.

If the norm of B were ultrametric, then $\|\omega\|_B \leq \max_j \|\omega_j\|_B = 1 < N/M_r$ for $N > M_r$. If $\|\cdot\|_u$ were an equivalent ultrametric norm, with $\|\cdot\|_B \leq c\|\cdot\|_u \leq c'\|\cdot\|_B$, then $\|\sum_{j < N} \omega_j\|_B \leq c \max_j \|\omega_j\|_u \leq c'$ for every N , contradicting $N/M_r \rightarrow \infty$. (Only the ultrametric inequality of $\|\cdot\|_u$ is used, not its homogeneity.)

For F : $\iota_F: B \rightarrow F$ is a K_1 -linear isometry (Lemma 35(3)), so the vectors $\iota_F \omega_j$ satisfy $\|\iota_F \omega_j\| = 1$ and $\|\sum_{j < N} \iota_F \omega_j\| \geq N/M_r$, and the same argument applies. Regarding F over a subfield does not change its norm. $\square$

If $r \leq \frac{1}{2}$, then $M_r = 1$ and $\|\sum_{j < N} \omega_j\|_B = N$ exactly.

6.2. Proof of Theorem 1. Let K be a nontrivially normed field of characteristic $p > 0$, and $k \geq p$.

- (1) *The base.* K is ultrametric (Lemma 4), and its completion $\widehat{K}$ is a complete, nontrivially normed, ultrametric field of characteristic p . Choose $t \in K$ with $0 < |t| = r < 1$; it exists because K is nontrivially normed.
- (2) *The Laurent subfield.* Let $K_1 \subseteq \widehat{K}$ be the closure of $\mathbb{F}_p(t)$. By Lemma 30 with $K' = \widehat{K}$, it is a closed subfield, isometrically isomorphic to $\mathbb{F}_p((X))$ with $|X| = r$ via ev_t , and it is complete and spherically complete.
- (3) *No lift over K_1 .* Put $E_1 := \ell^\infty(\mathbb{N}, K_1)$, and let B be the completion of $\Lambda_{K_1}^k E_1$ under the projective exterior norm. Theorem 26 with $\kappa = \mathbb{F}_p$ holds for $\mathbb{F}_p((X))$, and it transports

along the isometric isomorphism ev_t : via ev_t^{-1} a normed $\mathbb{F}_p((X))$ -space is a normed K_1 -space with the same group and norm, $\ell^\infty(\mathbb{N}, \mathbb{F}_p((X))) \cong \ell^\infty(\mathbb{N}, K_1)$ isometrically, and this identification preserves exterior powers, projective norms, the completion B , the spaces $\mathcal{L}$ and Alt^k , the maps Q and their bounded lifts. So $Q_{E_1, E_1, B}^{K_1}$ has no bounded k -linear lift over K_1 .

- (4) *Ascent to $\widehat{K}$* . Hypothesis (H1) holds for $K_1 \subseteq \widehat{K}$ by (2), since $|t| = r \in (0, 1)$, and (H2) holds by (1). The space $F_1 := B$ is complete. By Proposition 38 with $K' = \widehat{K}$, the map $Q_{E, E, F}^{\widehat{K}}$ for

$$E := E_1 \widehat{\otimes}_\pi \widehat{K}, \quad F := B \widehat{\otimes}_\pi \widehat{K}$$

has no bounded k -linear lift over $\widehat{K}$.

- (5) *Descent to K* . Regard E and F as K -Banach spaces. By Lemma 40, a bounded k -linear lift of $Q_{E, E, F}^K$ over K would be a bounded k -linear lift of $Q_{E, E, F}^{\widehat{K}}$ over $\widehat{K}$. So there is none.
- (6) *Nowhere analytic*. By Proposition 6, which holds over the possibly incomplete field K , $Q_{E, E, F}^K$ is analytic at no point, and $\text{ContDiffAt } K \ \omega$ fails at every point. For a finite set ι with $|\iota| = k$ the same holds, by the identification of §2.
- (7) *The target*. Since $k \geq p \geq 2$, Proposition 41 (with $\kappa = \mathbb{F}_p$, transported along ev_t as in step (3), and $K' = \widehat{K}$, regarding F over K) shows that F has no equivalent ultrametric norm. $\square$

Remark 42. If $K_1 = \widehat{K}$, for example if $K = \mathbb{F}_p((u))$ and $t = u$, then $E = E_1$ and $F = B$ (Lemma 35(3) and $x \otimes \lambda = \lambda x \otimes 1$), so E is ultrametric. For $K = \mathbb{F}_q((u))$, $q = p^n$ with the u -adic absolute value, one can also bypass the ascent: Theorem 26 with $\kappa = \mathbb{F}_q$ and $X = u$ gives $E = \ell^\infty(\mathbb{N}, K)$ and $F = B$ directly. We do not claim that E is ultrametric in general; the proof never needs it to be.

6.3. Proof of Corollary 2. Put $k := |\iota|$.

- (1) Assume $k! \neq 0$ in K , and let E, E', F be normed. Define

$$P(f_1, \dots, f_k)(m)(x_1, \dots, x_k) := \frac{1}{k!} \sum_{\sigma \in S_k} m(f_{\sigma(1)}x_1, \dots, f_{\sigma(k)}x_k).$$

For fixed f and m this is k -linear in x ; each of the $k!$ terms of the sum has norm at most $\|m\| \prod_i \|f_i\| \prod_j \|x_j\|$, so it is bounded. It is alternating: if $x_a = x_b = y$ with $a \neq b$, let τ be the transposition of a and b ; the terms for σ and $\sigma \circ \tau$ are $m(\dots, f_{\sigma(a)}y, \dots, f_{\sigma(b)}y, \dots)$ and $m(\dots, f_{\sigma(b)}y, \dots, f_{\sigma(a)}y, \dots)$ (slots a and b shown), which cancel because m is antisymmetric, and $\sigma \mapsto \sigma \circ \tau$ is a fixed-point-free involution of S_k . So $P(f)(m) \in \text{Alt}^k(E; F)$; it is linear in m , and P is a bounded k -linear map. On the diagonal all $k!$ terms equal $m(fx_1, \dots, fx_k)$, so $P(f, \dots, f) = Q(f)$. By Proposition 6(1), Q is given at every point by a finite power series; for $\iota = \emptyset$, where Proposition 6 assumes $k \geq 1$, this is trivial, because Q is then constant. No completeness is used.

- (2) Assume $k! = 0$ in K . In characteristic 0 the positive integer $k!$ is nonzero in K , so $\text{char } K = p > 0$, and $k! = 0$ in $\mathbb{F}_p$ means $p \leq k$. Theorem 1 gives the required spaces.

Together, (1) and (2) give the equivalence. $\square$

7. COMPLEMENTS AND OPEN PROBLEMS

Theorem A and targets over discretely valued bases. The following positive result is the third item of §1. A special case is stated without proof in [6, Remark 16]. We quote the general statement from an unpublished Lean formalization (§8) and do not prove it. It is used only in this subsection, and not in the proofs of Theorem 1 and Corollary 2.

Theorem A ([6, Remark 16] for a special case; formalized, §8). Let K be an ultrametric nontrivially normed field, let E, E' be normed K -spaces, and let F be a normed K -space whose norm is ultrametric and which is spherically complete as a metric space. Then $Q_{E, E', F}^K$ is C^n for every $n \in \mathbb{N} \cup \{\infty, \omega\}$; in particular it is analytic at every point.

Proposition 43 (targets over discretely valued bases). *Assume Theorem A. Let K be an ultrametric normed field with $|K^\times| = r^\mathbb{Z}$ for some $r \in (0, 1)$, complete or not; for instance $K = \mathbb{F}_q((t))$ or $K = \kappa((t))$. Let E, E' be normed K -spaces and F a K -Banach space that admits an equivalent ultrametric norm. Then $Q_{E, E', F}^K$ is analytic at every point.*

Proof. Rounding. Let $\|\cdot\|_u$ be an ultrametric norm on F equivalent to its given norm. For $y \neq 0$ let $\|y\|_u^\sharp$ be the least element of $\{s \in r^\mathbb{Z} : s \geq \|y\|_u\}$, which exists because $\|y\|_u > 0$, and put $\|0\|_u^\sharp := 0$. Then $\|y\|_u \leq \|y\|_u^\sharp \leq r^{-1}\|y\|_u$. Rounding up to $r^\mathbb{Z}$ is monotone and commutes with multiplication by elements of $r^\mathbb{Z}$. Since $|\lambda| \in r^\mathbb{Z}$ for $\lambda \in K^\times$, this gives $\|\lambda y\|_u^\sharp = |\lambda|\|y\|_u^\sharp$, and $\|y + y'\|_u^\sharp \leq \max(\|y\|_u^\sharp, \|y'\|_u^\sharp)$ follows from the same inequality for $\|\cdot\|_u$. So $\|\cdot\|_u^\sharp$ is an ultrametric norm on F , equivalent to the given one. Hence $F' := (F, \|\cdot\|_u^\sharp)$ is complete, and its distances lie in $r^\mathbb{Z} \cup \{0\}$.

Theorem A. By Lemma 28, F' is spherically complete, so $Q' := Q_{E, E', F'}^K$ is analytic at every point by Theorem A.

Equivalent norms. Boundedness and alternation of a multilinear map do not depend on the choice between equivalent norms on the target. So $\text{Alt}^k(E'; F) = \text{Alt}^k(E'; F')$ and $\text{Alt}^k(E; F) = \text{Alt}^k(E; F')$ as vector spaces, with equivalent operator norms. Hence $Z := \mathcal{L}(\text{Alt}^k(E'; F), \text{Alt}^k(E; F))$ and $Z' := \mathcal{L}(\text{Alt}^k(E'; F'), \text{Alt}^k(E; F'))$ are the same vector space with equivalent norms, and $Q = Q'$ as maps from $\mathcal{L}(E, E')$ to it. A power series of Q' at a point, with radius R and $\sup_n \|p_n\| R^n < \infty$ for the norm of Z' , is a power series of Q at that point with the same radius: the operator norms of the p_n change by at most a constant factor, and convergence in Z' is convergence in Z . $\square$

Consequently, over such K , the target F of every Banach counterexample (E, E', F) admits no equivalent ultrametric norm, as the target of Theorem 1 does (Proposition 41).

Remark 44 (necessary, but not sufficient). Over discretely valued bases, having no equivalent ultrametric norm is therefore necessary for the target of a Banach counterexample. The completeness of F is used in Proposition 43 through Lemma 28, and the statement concerns complete targets only. The condition is not sufficient. For $(p, k) = (5, 3)$, the space B of §4.2, built in degree 3 over $K_1 = \mathbb{F}_5((X))$, has no equivalent ultrametric norm by Proposition 41, which needs only $k \geq 2$. Yet $Q_{E_1, E_1, B}^{K_1}$ is analytic by Corollary 2(1), since $3! = 1$ in $\mathbb{F}_5$. The obstruction is the vanishing of $k!$. Indeed, feeding the bounded lift of Corollary 2(1) into §4.4 with $\kappa = \mathbb{F}_5$ produces a map Ψ with the properties of Lemma 25, which hold in every degree. So Ψ satisfies all the hypotheses of Theorem 15(2) with $L = \mathbb{F}_5$ except $k! = 0$; compare Remark 21.

Remark 45 (an explicit lift). Let K be ultrametric, let E have a basis $(e_i)_{i \in I}$ that is orthogonal for the norm (every x is $\sum_i e_i^*(x)e_i$, and $\|\sum_i c_i e_i\| = \sup_i |c_i| \|e_i\|$), and let F be complete and

ultrametric. Fix a total order on I and put

$$P(f_1, \dots, f_k)(m)(x^1, \dots, x^k) := \sum_{s_1 < \dots < s_k} \det(e_{s_a}^*(x^b))_{a,b} m(f_1 e_{s_1}, \dots, f_k e_{s_k}).$$

For every $\varepsilon > 0$ only finitely many terms have norm at least ε , so the family is summable in the complete ultrametric F . By orthogonality $|e_s^*(x)| \|e_s\| \leq \|x\|$, so by the ultrametric inequality in K every term has norm at most $\|m\| \prod_a \|f_a\| \prod_b \|x^b\|$, and $\|P\| \leq 1$ by the ultrametric inequality in F . The output is alternating because of the determinant; and on the diagonal the expansion $m(fx^1, \dots, fx^k) = \sum_{s_1 < \dots < s_k} \det(e_{s_a}^*(x^b)) m(fe_{s_1}, \dots, fe_{s_k})$ of an alternating map shows $P(f, \dots, f) = Q(f)$. This *sorted lift* is a bounded lift in every characteristic and degree, with no division by $k!$ and no spherical completeness; we do not use it. Over a complete discretely valued field every ultrametric Banach space is topologically isomorphic to a space $c_0(I)$ of null families [9, §1, Prop. 1 and its Corollaire], so it has such a basis for an equivalent norm.

Remark 46 (ultrametric targets over bases that are not spherically complete; announced). We state the following without proof. It is not proved in this paper, and nothing here depends on it. Let $k \geq 2$, and let K be a complete nontrivially normed field with $k! = 0$ in K (so K is ultrametric, by Lemma 4) that is not spherically complete. For a cardinal μ put $\beth_0(\mu) := \mu$ and $\beth_{n+1}(\mu) := 2^{\beth_n(\mu)}$, and let μ^+ be the successor cardinal of μ . Let I be a set of cardinality at least $\beth_{3k-1}(|K|)^+$, put $E := \ell^\infty(I, K)$, and let F_I be the closure in $\ell^\infty(I^k, K)$ of the span of the determinant arrays $(i_1, \dots, i_k) \mapsto \det(x_b(i_a))_{a,b}$, $x \in E^k$. Then E and F_I are ultrametric K -Banach spaces, and Q_{E,E,F_I}^K is analytic at no point. Our argument for this, which we do not give, is a partition argument with the Erdős–Rado theorem in place of Ramsey’s theorem. For $k = 2$ in characteristic 2 the countable index set $\mathbb{N}$ suffices [6].

Comparison with characteristic 2. For $p = k = 2$, Theorem 1 gives a second counterexample, with a non-ultrametric target, over *every* field of characteristic 2. This includes spherically complete fields such as $\mathbb{F}_2((t))$, where the hypothesis of the construction of [6], that the base field is not spherically complete, fails. Moreover, by Proposition 43, which rests on Theorem A and hence on an unpublished formalization, no counterexample over $\mathbb{F}_2((t))$ can have a complete ultrametric target such as that of [6]. Conversely, the target of [6] is ultrametric. The target of Theorem 1 is not, and it has no equivalent ultrametric norm (Proposition 41); over discretely valued bases no complete counterexample target can have one (Proposition 43). The mechanisms are different. The characteristic-2 argument is a parity identity for determinants (the determinant of an alternating matrix is the square of its Pfaffian), glued along ℓ^∞ through the failure of the Hahn–Banach theorem over fields that are not spherically complete. Here the finite-field theorem is combined with the constant-coefficient map of a Laurent series field, and Ingleton’s theorem is used in the opposite direction, over a spherically complete subfield. In characteristic 2, Theorem 15 needs no signs (Remark 20).

Open problems. All remaining questions concern ultrametric targets.

- (1) *Countable index.* Let K be complete, not spherically complete, of characteristic $p \geq 3$, and $k \geq p$. Is there a complete ultrametric F such that $Q_{E,E,F}^K$ is not analytic for $E = \ell^\infty(\mathbb{N}, K)$? For $p = 2$ the answer is yes, by [6]: for $k = 2$ this is the theorem of [6], and for $k \geq 3$ it follows by the wedging argument of [6, Remark 18], which is formalized in the Lean theorem of §8 under the hypothesis (K2) of [6]. The statement announced in Remark 46 would give such an F only for a much larger index set.
- (2) *Densely valued spherically complete bases.* Let K be spherically complete, densely valued and of characteristic p , for example the field $\mathbb{F}_p((t^\mathbb{Q}))$ of Hahn series, and let

$k \geq p$. Is $Q_{E,E',F}^K$ analytic for all Banach E, E' and every complete ultrametric F ? It is when F is spherically complete (Theorem A), and it is not for some non-ultrametric Banach F (Theorem 1).

- (3) *Formalization.* Theorem 1 is not machine-checked. Its most combinatorial part, Theorem 15, uses only finite combinatorics and Ramsey's theorem, and is a natural first target.

8. FORMALIZATION

The following statements are formalized in Lean 4 with Mathlib. They compile with no `sorry`, and the only axioms they use are `propext`, `Quot.sound` and `Classical.choice`. Apart from the pull request [7], the Lean files are not public; the repository of [6] is private. The Lean names below are those of this unpublished development (the namespace `Round24Transfer` is an internal name).

- The C^n statement [7]: `ContinuousAlternatingMap.contDiff_compContinuousLinearMapCLM`, for every $n : \mathbb{N} \cup \{\infty\}$, on the branch of the pull request.
- The characteristic-2 counterexample [6], at the point 0 and $|\iota| = 2$: the theorem `AltCounter.not_analyticAt_Q_of_NSC` of [6], whose hypothesis `NSC` is obtained from (K2) by `AltCounter.nsc_of_hasEmptyNestedBalls`. Its extension to every $|\iota| \geq 2$ and every base point, by the wedging argument of [6, Remark 18]: `Round24Transfer.char2_not_analyticAt_of_hasEmptyNestedBalls`. In both, the base field is assumed complete, ultrametric and of characteristic 2, and to satisfy the hypothesis (K2) of [6] (`AltCounter.HasEmptyNestedBalls`): there are closed balls $\mathcal{D}(c_n, r_n)$ with $|c_{n+1} - c_n| \leq r_n$, $r_0 > r_1 > \dots$ and empty intersection. By Remark 29, this is equivalent to the base field not being spherically complete.
- The lift criterion, Proposition 6, as the six-fold equivalence `Round24Transfer.tfae_Q` (bounded lift; continuously polynomial at every point; analytic at every point; analytic at some point; C^ω ; C^ω at some point), for all normed spaces over any nontrivially normed field (the infinite radius in Proposition 6(1) is proved on paper).
- The positive direction of Corollary 2: `ContinuousAlternatingMap.cpolynomialAt_compContinuousLinearMapCLM` and `ContinuousAlternatingMap.contDiff_compContinuousLinearMapCLM_of_factorial_ne_zero`, under the hypothesis $((\text{card } \iota)! : K) \neq 0$, for all normed E, E', F .
- Theorem A (§7): `ContinuousAlternatingMap.contDiff_compContinuousLinearMapCLM_of_sphericallyComplete`.
- Auxiliary results of this paper: Lemma 4; the case of Lemma 28 in which Y is a normed field, which gives Lemma 30(4); and a form of Ingleton's theorem (Theorem 32) with a spherically complete ultrametric codomain.

Theorem A and the last two auxiliary facts use a class of spherically complete spaces that is defined in the unpublished development, with the definition of §5; it is not part of Mathlib. Theorem 15, Proposition 24, Lemma 25, Theorem 26 and the base change of §5 are not formalized, and hence neither are Theorem 1 and the negative direction of Corollary 2. Proposition 43 is not formalized either, apart from Theorem A, which it uses.

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